ZEROGUARD: A Solver-Free Battery Safety Certificate for Autonomous Vehicles, from Robotaxis to Deep Space

Anchored Collapse and the geometry of the admissible input set

Abstract

An autonomous vehicle's battery must be pushed hard enough to be useful and held back enough to be safe, and the state deciding where that line falls — series resistance, plating margin, true capacity — is not measurable in the field. Every unattended vehicle inherits the same compromise: a fixed de-rate chosen offline for the worst cell at the worst temperature at end of life, applied whatever the present state happens to be. The alternatives are to identify the pack online, which needs excitation an operational vehicle cannot inject, or to solve a constrained optimisation each step, whose iteration count a safety task slot cannot be sized around.

We prove that neither is necessary. Terminal voltage, temperature and the lithium-plating potential are each *monotone* in applied current, so the admissible input set is a single **interval** whose edges are found by bisection in a *fixed* number of one-step model evaluations — no optimiser, no convergence test, no identification. The published form assumes the safe input is zero, which holds for a cell being charged and fails for any vehicle doing its job. That failure is a misdiagnosis: zero still satisfies every temperature and voltage constraint on a discharging pack; what zero cannot do is serve the load, which is a *floor* rather than a cap. **Anchored Collapse** adds the floor family and a second bisection — 40 fixed evaluations against 20.

What that buys is not speed but *transfer*. A de-rate tuned on a fleet a manufacturer could characterise in advance violates in 26.2 % of deployment sessions; the certificate, re-tuned not at all because it has nothing to tune, violates in 0 of 400. And against the control-barrier-function quadratic program — the standard safety filter, and the method's own family — the relationship turns out to be closer than rivalry: enforced on the true model rather than a linearisation, the discrete-time CBF condition for a monotone constraint *is* an interval, so this is the bisection that solves that problem exactly, with γ entering as a margin.

Three things make the guarantee checkable rather than asserted. It is **machine-checked in Lean 4** — nine theorems, no `sorry`, no dependency on any axiom — and the proof needs less than we claimed: safety holds for an *arbitrary* predicate, so monotonicity buys optimality, not safety. The two assumptions it consumes are tested against 33 cells measured by another laboratory, and the filter itself against a Doyle–Fuller–Newman plant, a different model class, where the affine plating proxy kept the DFN's *actual* electrode potential above the deposition onset in every session. And it ships: 2,064 B of compiled code, 68 B of RAM, worst-case 123.9 μs — 1.24 % of a 10 ms ISO 26262 slot.

Across 16 platforms in four media, 61 experiments and 506,747 episodes, nothing was breached while the certificate reported feasible. **Twenty of those experiments contradicted the hypothesis they were written to test** and are reported as findings — among them the two media of four where the incumbent rule is *not* beaten, which the method predicts in advance from whether mission closure is envelope- or energy-driven.

1 Introduction

The limiting resource for long-duration autonomy is not perception, planning, or control. It is *energy under certainty*: how hard a vehicle may push its battery when nobody is available to inspect it, and how confident anyone can be in the answer.

The gap between what a lithium-ion pack can physically deliver and what a battery management system will actually permit is large, and it is not there by accident. Fast charging drives three coupled failure modes — thermal runaway, over-potential, and lithium plating on the anode — whose onset depends on internal states that a production sensor suite does not measure. Series resistance roughly doubles over a pack's service life, plating risk climbs as temperature falls, and the safe current at end of life in cold weather is a fraction of the safe current when new and warm. A manufacturer facing that uncertainty does the only defensible thing and de-rates for the worst case, permanently.

For a privately owned car this is a mild inefficiency. For an autonomous fleet it is the economics of the whole enterprise. A robotaxi that charges twelve times a day accumulates a private car's decade of fast-charge stress in 3.0 weeks (§6); the de-rate it inherits was designed for a vehicle with a fundamentally different duty cycle. For a spacecraft it is worse still, because the de-rate is chosen before launch and can never be revisited.

1.1 Contributions

1. **A geometric fact, and its consequence.** Voltage, temperature and plating potential are monotone in current, so the admissible input set is an *interval*. Its edges are found by bisection at a cost fixed before run time: 20 model evaluations on charge, 40 on discharge. No optimiser, no convergence test, no identified model (§4).
2. **Anchored Collapse.** The published theorem assumes the safe input is zero, which is true of a cell on a bench and false of any vehicle serving a load. Splitting the constraints into caps and floors and running a second bisection removes that assumption, and is what carries the result from a charger to a rotorcraft, an AUV and a spacecraft (§4).
3. **It is the exact CBF filter, not a rival to it.** Enforced on the true model rather than a linearisation, the discrete-time control-barrier-function condition for a monotone constraint defines an interval. This work is the bisection that solves it exactly, with γ entering as a margin; the quadratic program people deploy is its linearised approximation (§13).
4. **Machine-checked, and it needed less than we claimed.** nine theorems in Lean 4, no `sorry` and no axiom. Writing them showed that safety holds for an *arbitrary* predicate: monotonicity buys optimality, not safety (§4.3).
5. **The guarantee transfers where a tuned constant does not.** Held to a tune-then-deploy standard, the incumbent rule fails in two of four media and the certificate in none — and in the other two the incumbent is *not* beaten, which §21 predicts in advance from whether mission closure is envelope- or energy-driven (§12, §22.1).
6. **Checked against physics we did not write, and a compiler.** The assumptions against 33 cells from another laboratory; the filter against a Doyle–Fuller–Newman plant, where the affine plating proxy held the *actual* electrode potential above the deposition onset; and the implementation as 2,064 B of compiled C agreeing with the reference at all 20,000 states tested (§16, §17, §13.1).
7. **A register that was not written to pass.** 61 experiments, 506,747 episodes; twenty of them contradicted the hypothesis they were written to test and are reported as findings rather than removed (Appendix).

1.2 Two standard answers, and why neither reaches the vehicle

The first is *identification*: estimate the pack's internal state online and de-rate against the estimate. Estimating resistance and capacity well requires excitation — current profiles rich enough to be informative — which an operational vehicle is not free to inject. Worse, the guarantee becomes conditional on the estimator having converged, which is a claim about a transient nobody audits.

The second is *constrained optimisation*: solve a small model-predictive control problem each step and let the solver enforce the constraints. This is the standard modern answer, and a good one, with a specific defect — a quadratic program's iteration count depends on its data. A safety-critical task slot is sized for the worst case, and if the worst case is unbounded the slot cannot be sized at all. In practice this is handled with iteration caps and fallback logic: by having a plan for when the guarantee does not hold.

1.3 The result

We take a third path, whose value is that it needs neither. The observation is geometric rather than numerical:

> If every constraint is monotone in a scalar input, and one input is known to be safe, then the set of admissible inputs is not a complicated region to be searched. It is a single interval, and its edges can be located by bisection in a fixed number of one-step model evaluations.

Bisection needs no optimiser, no convergence test, and — critically — no identification of the plant, because a monotone *bound* on the uncertain parameter suffices where a point estimate would be required. A datasheet end-of-life resistance figure is such a bound and is known before the vehicle ships.

The contribution of this paper is to establish that this result survives contact with real vehicles, which required repairing it first. Section 4 shows that the failure observed when the filter is placed on a hovering rotorcraft is not the failure it appears to be, and develops **Anchored Collapse**, which covers discharge as well as charge. Sections 6–9 then take it across four media — ground, air, water, vacuum — across 16 platforms and 61 experiments.

Contributions. **(i)** *Anchored Collapse* (§4.2), a generalisation covering constraints requiring the input to be *large* enough — what every vehicle's irreducible load imposes and what the original theorem cannot express — at a cost of 40 fixed evaluations rather than 20. **(ii)** A two-sided certificate and a *reserve* (§4.5): the gap between the edges is how much room the vehicle has left, which the one-sided version cannot report and which yields an abort rule for vehicles that cannot afford to find their limit empirically. **(iii)** A certificate that refuses *designs*, not just commands (§7.1). **(iv)** Application across four media (§6–§9), including where the result does *not* hold and what that costs.

2 The autonomy energy bottleneck

Why a battery certificate is a bottleneck for autonomy rather than a component-level concern is worth being concrete about, because the four application sections that follow are chosen to span the ways it binds.

Ground fleets are duty-cycle limited. A robotaxi's economics are set by utilisation, and utilisation is set by how long it spends attached to a charger. The constraint is not the charger — §6 shows a 350 kW station is already past what the pack will accept — but how aggressively the pack may be driven at the temperature it happens to be, at the age it happens to be, with no technician in the loop. That is precisely the quantity a certificate computes.

Aerial autonomy is margin limited. A rotorcraft in hover draws a continuous 1–3C from its pack. There is no coasting, no regeneration, and no graceful degradation: an aircraft that runs out of admissible current does not slow down, it descends. Every kilogram of battery carried to buy margin is a kilogram not carried as payload, so the margin is contested directly against the mission.

Underwater autonomy is abort limited. An AUV under an ice shelf has no abort-to-surface. The vehicle must decide to turn back while turning back is still possible, using only what it can compute onboard, hours or days from any human. A certificate that is correct but late is indistinguishable from one that is wrong.

Space autonomy is calibration limited. A spacecraft's battery model is fixed at integration and cannot be updated in any meaningful sense afterwards — the downlink budget is not for battery parameters, and the round-trip light time to the outer planets is hours. Over a five-year mission the pack ages, takes radiation dose, and cycles tens of thousands of times against a parameter set that has not changed since before launch. **The property that makes a solver-free certificate merely convenient on a car is the property that makes it admissible at all on a spacecraft**: it requires no identification, so there is nothing to recalibrate.

These four are not a list of markets. They are four different ways for the same underlying guarantee to be the binding constraint, and a method that addresses one by exploiting its specifics would not transfer. What follows is deliberately one method, unmodified, in all four.

2.1 Related work

The framework is control invariance [1]: a set is control-invariant if from every state in it some input keeps the next state inside. Control barrier functions [2] and predictive safety filters [3] enforce this by solving an optimisation each step; reference governors [4] and shielding [5] are related. §13 measures against the CBF-QP directly, and finds the relationship is closer than rivalry: enforced on the true model rather than a linearisation, the discrete-time CBF condition for a monotone constraint defines an interval, and this method is the bisection that solves it exactly. Ours is a predictive safety filter whose optimisation has been replaced by a bisection, which is possible only because of monotonicity [6; 7] — order-preserving structure that lets worst-case reachability be computed from extreme points rather than by searching an interior. Our use is narrower, monotonicity in a scalar *input* rather than in the state, and it is what turns a search into two bisections.

The reference battery model is Doyle–Fuller–Newman [8; 9] with Bernardi thermal coupling [10], in PyBaMM [11] with the Chen [12] and O'Kane [13] parameter sets; plating is the dominant fast-charge degradation mode [14]. Online estimation of resistance and capacity is mature [15; 16] — and is exactly what this method deliberately does not require. Reinforcement learning has been applied to fast charging [17; 18], and constrained RL offers expectation-level guarantees [19; 20], which are the wrong shape for a constraint that must hold on every step of every episode. Our position is that the policy may be learned and the safety must not be.

3 The cell, and which of its constraints are monotone

The result of §4 is a statement about constraints, so it is worth writing the constraints down. Everything in this paper rests on a reduced-order electro-thermal model of a lithium-ion cell, calibrated against a Doyle–Fuller–Newman simulation in PyBaMM and reported in [21] at a held-out voltage

RMSE of 20.3 mV and temperature RMSE of 1.32 °C across ambients from 0 to 25 °C. The model is prior work; what matters here is its *shape*, because that is what the theorem consumes.

State and outputs. The state is $x = (\mathrm{soc}, T, V_1)$ — state of charge, cell temperature, and the overpotential stored in a single RC branch — with a slowly varying ageing pair (capacity loss, resistance growth) carried alongside. For an applied current I (positive charging) over a step Δt, the model returns terminal voltage, the next temperature, and a plating-potential proxy:

$$V = \mathrm{OCV}(\mathrm{soc}) \; + \; \underbrace{I\,R_\Omega(\mathrm{soc}, T)}_{\text{ohmic}} \; + \; \underbrace{A\big(1 + c_T(T-25)\big)\,\mathrm{asinh}\big(I/2i_0(\mathrm{soc})\big)}_{\text{activation}} \; + V_1, \tag{1}$$

$$C_{\mathrm{th}}\,\dot{T} = \underbrace{I\,\eta_\Omega + I\,\eta_{\mathrm{act}} + V_1^2/R_1}_{\text{irreversible}} \; + \; \underbrace{I\,(T + 273.15)\,\partial U/\partial T}_{\text{reversible}} \; - q_{\mathrm{out}}(T), \tag{2}$$

$$\varphi_{\mathrm{an}} = p_0(\mathrm{soc}) \; - \; p_1(\mathrm{soc}, T)\,I, \qquad p_1 > 0. \tag{3}$$

R_Ω carries an Arrhenius temperature dependence, so resistance rises as the cell cools; q_{out} is the cooling law, and it is the *only* thing that changes between the four media of §6–§9.

Why each constraint is monotone in current. This is the property the theorem needs, and each one has a physical reason rather than an empirical one.

- **Voltage** (1). Both overpotential terms are odd and increasing in I (asinh is strictly increasing), and OCV does not depend on I. So V increases in charge current and decreases in discharge current: a cap on charge, a cap on discharge as well, because the constraint there is $V \geq V_{\min}$ on a signal that falls.
- **Temperature** (2). Every generation term is non-decreasing in $|I|$ — $I\eta_\Omega = I^2 R_\Omega \geq 0$ regardless of sign — and q_{out} does not depend on I at all. So the next temperature is non-decreasing in current, whatever the cooling law, which is why replacing Newtonian convection with Stefan–Boltzmann radiation (§9) leaves the certificate's structure untouched.
- **Plating** (3) is *affine* and strictly decreasing in I. Plating occurs when $\varphi_{\mathrm{an}} \leq 0$.
- **Delivered power** $P = S\,V(I)\,I$ is increasing in I below the maximum-power point — the one condition of Remark 2.

Why plating is enforced and not certified. This distinction recurs throughout the paper and it is a consequence of (3) rather than a modelling choice. Forward invariance requires an always-available input that returns the state to the safe set. At $I = 0$, $\varphi_{\mathrm{an}} = p_0(\mathrm{soc})$, and p_0 falls with state of charge: above roughly 0.82 SOC it is already below the required margin. **There is no current — not even zero — that clears the plating margin at high state of charge**, so no one-step condition can render it invariant, and no honest certificate can claim it.

The certified set is therefore $\mathcal{S} = \{T \leq T_{\max},\ V \leq V_{\max}\}$, and plating is *enforced* by the same bisection plus a temperature-dependent current cap, and monitored. It rides the same monotone structure, so it costs nothing extra to enforce; it simply cannot be promised. §8 is where that distinction stops being bookkeeping and becomes the binding constraint of an entire vehicle class.

4 Theory

Let a system have state x, scalar input $u \in [u_{\min}, u_{\max}]$, one-step map $x^+ = f(x, u, w)$, and a finite family of constraints $c_j(x, u) \leq 0$ defining a safe set $\mathcal{S}$. Write $\mathcal{U}(x) = \{u : c_j(x, u) \leq 0\ \forall j\}$ for the

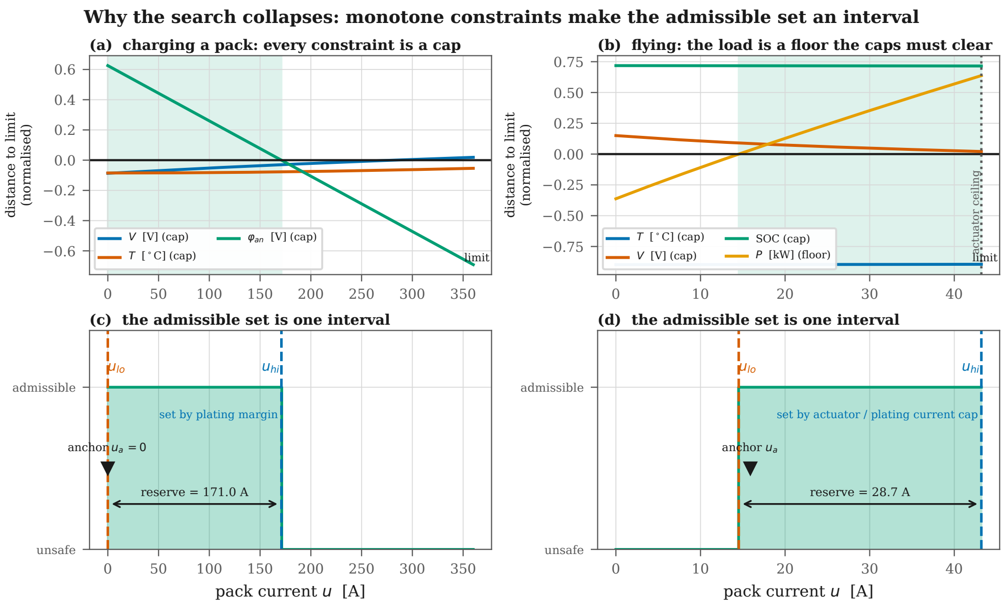

Figure 1: Why the search collapses, drawn from the model of (1)–(3) at one fixed state rather than sketched. **(a)** Charging: every constrained signal is a cap, each crossing its limit exactly once, and the admissible set runs from zero to whichever crosses first — here the plating margin at 171 A, well before voltage at 281 A. **(b)** Flying: the caps still come down from above, but the load's power requirement comes *up* from below, and the certificate is the gap between them. **(c,d)** The resulting admissible set: one interval, with the anchor inside it, and a width that is the reserve.

admissible input set.

4.1 Null-Input Collapse

Proposition 1 (Null-Input Collapse)**.** *Suppose **(A1)** $u = 0$ is safe — from every $x \in \mathcal{S}$ the one-step map at $u = 0$ returns to $\mathcal{S}$ — and **(A2)** each c_j is non-decreasing in u. Then $\mathcal{U}(x) = [0, u^*(x)]$ is a single interval anchored at zero, $\mathcal{S}$ is forward-invariant under any policy filtered through it, and u^* is computed to precision ε in $\lceil \log_2(u_{\max}/\varepsilon) \rceil + 2$ one-step evaluations.*

Sketch. By (A2) each $\{u : c_j \le 0\}$ is a down-set; a finite intersection of down-sets in $\mathbb{R}$ is a down-set, hence an interval $[u_{\min}, u^*]$, which by (A1) contains 0. Bisection on a monotone predicate converges linearly with an iteration count depending only on the requested precision, not on the data. Forward invariance follows by induction: each filtered step lands in $\mathcal{S}$ and $u = 0$ remains available. $\square$

Two things are worth isolating. First, no step requires knowing the plant's uncertain parameters — only evaluating c_j under a *bound* on them, and a conservative bound yields a conservative interval. Second, the cost is $O(\log 1/\varepsilon)$ *deterministic* evaluations, which is what a task slot can be sized around and a solver's iteration count is not.

4.2 Where it breaks, and why that reading is wrong

Placed on a quadrotor holding altitude, a filter built on Proposition 1 [21] passes every one-step check for a mean of 22.3 consecutive steps and then violates in 100 % of episodes. The natural reading is that (A1) has failed: $u = 0$ is plainly not safe for an aircraft.

That reading is wrong, and the correction is the technical heart of this paper. Consider what $u = 0$ actually does to a discharging pack. No current flows, so there is no I^2R heating and the temperature relaxes toward ambient. Terminal voltage returns to open-circuit, so it cannot sag below a floor. State of charge does not fall. **Every temperature and voltage constraint is satisfied at $u = 0$, on a quadrotor in flight exactly as on a cell being charged.**

What $u = 0$ cannot do is fly the aircraft. The constraint that failed is of a different kind: a requirement that the input be *large* enough. Call these *floors*, as against the *caps* of Proposition 1. Every vehicle has at least one, because every vehicle has an irreducible load — the power it cannot stop drawing and remain a vehicle. Hover thrust for a rotorcraft. Hotel load for a submersible, whose computer and inertial navigation draw their watts whether the thrusters do or not, so that even a stationary AUV cannot command zero. Bus load for a spacecraft in eclipse. Contracted export power for a car doing vehicle-to-grid.

Charging is the special case where the irreducible load is zero — which is exactly why the original filter could anchor at zero without ever noticing it was making a choice.

Theorem 1 (Anchored Collapse)**.** *Partition the constraints into caps $\mathcal{C}$, each satisfied on $[u_{\min}, u_c]$, and floors $\mathcal{F}$, each satisfied on $[u_c, u_{\max}]$. Suppose **(A1′)** $u = u_{\min}$ is safe for every cap, from every $x \in \mathcal{S}$, and **(A2)** each constraint is monotone in u. Then*

$$\mathcal{U}(x) \;=\; [\,u_{\mathrm{lo}}(x),\; u_{\mathrm{hi}}(x)\,] \;=\; \Big[\, \max_{j\in\mathcal{F}} u_{c_j},\;\; \min_{j\in\mathcal{C}} u_{c_j}\,\Big], \tag{4}$$

both edges are found by bisection in $O(\log 1/\varepsilon)$ one-step evaluations, and $\mathcal{U}(x) \neq \emptyset$ exactly when $u_{\mathrm{lo}} \le u_{\mathrm{hi}}$.

Sketch. Each cap's satisfying set is a down-set and each floor's an up-set; their finite intersections are $[u_{\min}, u_{\mathrm{hi}}]$ and $[u_{\mathrm{lo}}, u_{\max}]$, whose intersection is the stated interval. (A1′) places $u_{\min}$ inside the cap family, giving the upper bisection a known-good left endpoint; $u_{\max}$ is inside the floor family if anything is, giving the lower bisection its right endpoint. Neither bisection needs the load current as a seed. □

Proposition 1 is the case $\mathcal{F} = \emptyset$. Note that (A1′) is *strictly weaker* than (A1): it asks only that zero satisfy the caps, which is the part that survives contact with a vehicle.

Remark 1 (The side of a constraint is data, not inference)*.* The plating constraint is $\varphi_{\mathrm{an}} \ge \delta$ — a lower-bound test on a signal that *decreases* in current, so it caps current from above. Sense and side coincide only when the constrained signal increases in u. Inferring side from the operator would file that cap among the floors and let the filter command straight through it: silent, and unsafe. Plants declare the side; an undeclared constraint raises rather than defaults.

Remark 2 (A soundness condition that is not obvious)*.* Delivered bus power $P = S\,V(u)\,u$ with $\mathrm{d}V/\mathrm{d}u < 0$ peaks at the maximum-power point and *falls* beyond it, so past that peak the set where the load is met stops being an up-set and the lower bisection would search a non-monotone family. The voltage cap prevents it: a pack reaches its maximum-power point near half its open-circuit voltage, far below any usable $V_{\min}$, so $u_{\mathrm{hi}} < u_{\mathrm{MPP}}$ always and the floor search is bounded by u_{hi}. This ordering is checked on every discharge platform, not assumed.

4.3 Machine-checked

Theorem 1 and the soundness of the bisection are formalised in Lean 4 and checked mechanically (`formal/AnchoredCollapse.lean`). nine theorems compile with no `sorry`, and Lean reports that each **depends on no axioms at all** — not classical choice, not propositional extensionality.

The proofs turned out to need less than the paper assumes: they are stated over an arbitrary type carrying a $\leq$ relation, with no order axiom used anywhere, so they hold without transitivity, antisymmetry or totality.

Writing the proofs also *weakened* the method's hypotheses. The safety guarantee turns out never to touch the cap/floor structure at all: `bisect_sound_any_predicate` holds for an *arbitrary* predicate, monotone or not, connected or not. So (A2) buys **optimality, not safety** — a non-monotone constraint set costs delivered charge, because the bisection finds only the first admissible interval, and it cannot cost the guarantee. For anyone applying this to a plant whose constraint structure they have not verified, that is the difference between a method they must first prove things about and one they can simply run.

The property worth checking mechanically is not the collapse theorem but `bisect_sound`: *every* state the bisection can reach has a feasible lower endpoint. Convergence says how good the answer is; this says whether it is safe, and it holds independent of iteration count, of the midpoint rule, and of floating-point behaviour — the proof never assumes the midpoint is the arithmetic mean, only that it lies between the endpoints. A filter interrupted by a deadline, starved of its budget, or run with a sloppy division still returns an admissible input.

4.4 Cost

Two probes bracket the cap family, then eighteen halvings; two more bracket the floors, then eighteen more. Worst case 40 one-step model evaluations on discharge and 20 on charge, both fixed and both branch-bounded. Measured worst p99 latency across all 16 platforms is $123.9\,\mu$s (buoyancy glider), which is $1.24\,\%$ of a 10 ms ISO 26262 task slot. On the ground platforms the figure is $59.7\,\mu$s, or $0.597\,\%$.

The property that matters is not that the number is small. It is that it does not move with the data.

4.5 The reserve

The two-sided certificate makes available a quantity the one-sided version cannot express: the width $u_{\rm hi} - u_{\rm lo}$ shrinks to zero as the envelope closes, so the filter can report *how close to unrecoverable* a vehicle is before it arrives. We treat this claim adversarially in §7 and scope it in §21: it carries the most operational weight and is the most likely to be overstated.

5 Method and platforms

5.1 The filter

Given a proposed input $u_{\rm req}$, the filter returns $\mathrm{clip}(u_{\rm req}, u_{\rm lo}, u_{\rm hi})$, or reports the interval empty. Empty is not a violation: it is the certificate stating correctly that the irreducible load can no longer be served inside the envelope, which is the true statement and the one a mission planner needs.

Robustness enters through margins rather than estimation: a thermal throttle $\delta_T = \delta_{T0} + K(s_R - 1)$ scaled by the *assumed* resistance bound s_R, with K the smallest value certifying cells past the rated bound, and a cooling reserve f that certifies any cooling loss up to $f/(1+f)$.

Table 1: The 16 platforms. Charge-mode platforms have a zero anchor and test Proposition 1 at vehicle scale; discharge-mode platforms have a positive anchor and test Theorem 1.

Medium	**Cooling law**	**Platforms**	**Irreducible load**
Ground	Newtonian, liquid loop	robotaxi 96S45P 78 kWh; shuttle; haul truck 288S115P 601 kWh; V2G	0 (charge); 7.4–22 kW export
Air	forced air, density-derated	delivery quadrotor 6S3P; eVTOL air taxi 300S45P; turnaround	hover thrust; 400 kW; 0 (charge)
Water	sealed hull into 2 °C	survey AUV; buoyancy glider; under-ice AUV; dock recharge	25–265 W; 0.5 W; 175 W; 0 (charge)
Vacuum	Stefan–Boltzmann to 4 K (+ 6 mbar CO_2 on Mars)	LEO smallsat; GEO comsat; deep-space cruiser; Mars rover; lunar lander; 500 krad orbiter	60 W; 3 kW; 200 W; 100 W; 45 W heater; 0 (charge)

5.2 One cell, sixteen vehicles

Three things vary across the platforms and nothing else does: the **medium**, through the cooling law; the **pack**, through series/parallel counts; and the **load**, through the irreducible draw. Table 1 lists them.

The cell underneath is the reduced-order model of [21] in every case — same overpotential form, same plating proxy, same calibrated coefficients — with two changes the vehicles require. Current may be negative, because vehicles discharge. And the Newtonian cooling term is a callable, because a quadrotor at 400 m, an AUV at 2 °C and a spacecraft in vacuum do not share a cooling law. Under the Newtonian law with positive current the vehicle cell reproduces the published model *bit for bit*, and every result downstream inherits that.

Pack geometry, hover powers, hotel loads and orbit parameters are *class-representative* — placing each vehicle in the right region of its class's design space using published figures — and are not any manufacturer's design. The purpose is that "it works on autonomous vehicles" is not a testable sentence, while "it holds a 96S45P pack under a 30-minute 350 kW session at 35 °C ambient" is.

5.3 Experimental discipline

Every experiment runs a *pessimistic estimator against a drawn plant*: the filter is given a platform at the conservative corner of the parameter envelope, and the plant it drives is drawn from inside it. A filter tested against its own model is testing arithmetic. One property makes that corner well defined for a two-sided certificate — scaling the assumed resistance up lowers the upper edge *and* raises the lower edge, so the interval shrinks from both sides under one perturbation — and it is verified numerically, not assumed.

The experimental surface, including what would falsify each claim, was fixed in a design register before any code ran; divergences between the register and what happened are recorded rather than edited into agreement. Zero-failure claims are reported with Clopper–Pearson 95 % one-sided upper bounds [22]; continuous outcomes with bootstrap confidence intervals; paired comparisons with Wilcoxon signed-rank and effect sizes; monotonicity claims with permutation Spearman.

6 Ground: robotaxis, shuttles, autonomous trucks

The ground domain is where the original result should hold most directly — the anchor is zero, the medium is a liquid loop, the physics is what the model was calibrated on. What is new is scale and

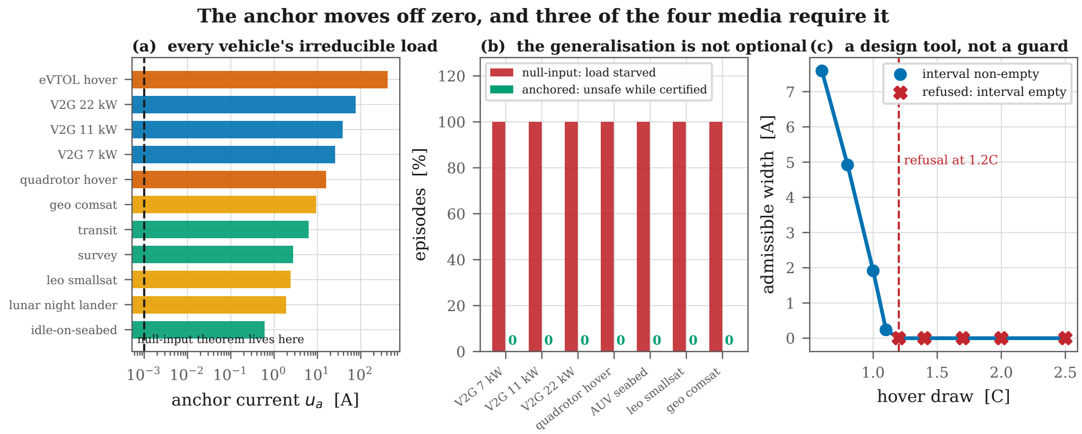

Figure 2: The anchor moves off zero, and three of the four media require it. (a) The irreducible load of every platform, spanning five orders of magnitude from a glider's 0.5 W hotel draw to an eVTOL's 400 kW hover; the null-input theorem lives at the dashed line. (b) On the same seeds, the null-input filter starves the load in essentially every episode while the anchored filter breaches nothing. (c) Applied before flight, the certificate refuses the calibrated energy cell for a rotorcraft above 1.2C of hover draw.

duty.

6.1 The filter is unchanged, at pack scale

On one cell the vehicle filter is bit-exact with the published projection: 0 mismatches in 100,000 states, across both discretisations, fresh and aged. A P-parallel pack's admissible current is exactly P times one cell's, to a relative deviation of 6.5×10^{-16}. And on a 33,120-cell haul-truck pack the admissible current equals the minimum over its heterogeneous cells to $1.9e-13$ A — below the $4.3e-13$ A resolution of the search itself. A pack-averaged thermal limit on that many cells is a statement about a cell that does not exist; the weakest-cell result is what makes a per-cell certificate tractable.

6.2 Megawatt charging and the weakest cell

The haul truck is the scale case, and it is where the per-cell architecture stops being a design preference. A 288S115P pack is 33,120 cells with individually different resistances, capacities and plating margins. A pack-averaged thermal limit on that many cells is a statement about a cell that does not exist — the average cell is not the one that fails.

The per-cell alternative is only tractable because of a structural result: the pack's admissible current is exactly the minimum over its cells'. We verify this from $N = 1$ to $N = 33,120$ with heterogeneous ageing, and the agreement is $1.9e-13$ A — below the $4.3e-13$ A resolution of the search itself, which is to say exact. The consequence is that a fleet operator does not need 33,120 filters; one filter evaluated at the weakest cell's parameters is provably the pack's answer.

Megawatt charging makes this urgent rather than academic. The binding constraint at that power is not the connector or the cable; it is the coldest, most-aged cell in a 600 kWh pack, and identifying which cell that is by measurement is not something a truck stop can do.

6.3 A 350 kW session

Under the full parameter envelope, 4,000 sessions on a 78 kWh robotaxi pack at 35 °C produced 0 violations (Clopper–Pearson 95 % upper bound 0.075 %), with a worst plant temperature of 39.9 °C against a 45 °C limit, and the pessimistic corner dominating every draw. Note which constraint binds: 350 kW into a 96S pack is about 985 A, above the pack's own 3C ceiling of 540 A. **The charger is not the limiting element; the certificate is.**

Over five years of robotaxi duty — 21,915 fast-charge sessions with the filter given a fixed datasheet bound on day one and told nothing afterwards — there were 0 violations (CP-95 0.0137 %). At twelve DC fast charges a day against a private car's one a fortnight, a factor of 171, a robotaxi accumulates a private car's *decade* of fast-charge stress in 3.0 weeks.

6.4 The result this domain was not looking for

Coupled straight to ambient air, the throttled margin puts the admissible ceiling at $T_{\max} - \delta_T = 32.5\,\check{r}C$. Above roughly that ambient there is *no* admissible fast charge at all, and the certificate delivers nothing — correctly, because at the datasheet end-of-life resistance bound there is none to deliver. The measured dead zone begins at 35 °C, the first grid node past the prediction.

Running the identical sweep with a loop holding coolant at 25 °C recovers the whole range and buys **+8.7 points of delivered charge** on average, with 0 violations in all 8,000 sessions across both configurations. That difference is exactly what a thermal management system purchases, quantified, and it is why production packs carry chillers rather than radiators.

A smaller finding follows. Delivered charge is not monotone in ambient and was never going to be: it peaks at 25 °C, rising along the cold limb ($\rho = +0.884$) where series resistance limits and falling along the hot limb ($\rho = -0.661$) where the thermal margin does. A *binary* service metric — "will it make 60 % in thirty minutes" — jumps the full range between adjacent grid cells, because thresholding a smooth quantity produces a threshold. Dispatch should plan on the curve the filter already computes, not on the binary derived from it.

6.5 What the certificate is worth to a fleet

These quantities translate into fleet operations directly, and the mechanism is the clearest case in this paper of a safety result having an economic consequence.

A production pack ships with a de-rate chosen for a worst case: end of life, temperature extreme, weakest cell. That worst case is almost never the present case. A vehicle that is warm, mid-life and mid-charge is being held to a limit designed for a vehicle that is none of those things. The certificate replaces the fixed de-rate with an evaluation at the *current* state under a bound that is still conservative — nothing is given up in safety — and the difference is recovered continuously.

The 8.7-point gain from active cooling above is one instance made measurable. So is the ambient curve: a dispatcher who knows delivered charge peaks at 25 °C and falls along each limb for a different physical reason can schedule charging around weather rather than against it. And the stress-equivalence result sharpens the whole question — a robotaxi is not a car that drives more, it is a vehicle whose battery ages at roughly 171 times the rate its de-rate was designed around. A guarantee holding for 21,915 sessions with no recalibration is what makes that duty cycle defensible rather than merely profitable.

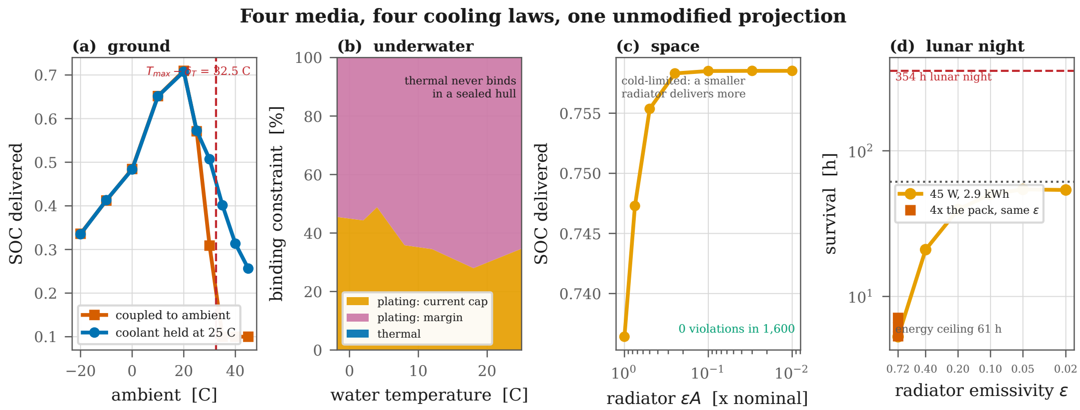

Figure 3: Four media, four cooling laws, one unmodified projection. (a) Ground: the passive ceiling at $T_{\max} - \delta_T$ and what a coolant loop recovers. (b) Underwater: plating binds at every water temperature and thermal never does; what shifts with cold is the share taken by the temperature-dependent current cap. (c) Space: delivered charge *rises* as the radiator shrinks, because the pack is cold-limited. (d) Lunar night: capacity barely moves survival, insulation moves it by an order of magnitude, until the energy ceiling takes over.

6.6 Sensing, transients, and vehicle-to-grid

A thermistor reading low is survivable up to a breakpoint with a closed form, $b^* = \delta_T/(1 - \Delta t\, hA/C_{\mathrm{th}}) = 12.802$ řC, using the throttled margin actually in force. The measured breakpoint is 12.9 °C on a 0.1 °C grid — an error of **0.098 °C**, from published constants with nothing fitted. Regenerative-braking pulses at 10 Hz over 1,500 episodes produced zero certified violations, and a coolant-degradation sweep over 8,000 trials violated nothing anywhere inside the region the cooling reserve predicts.

Finally, vehicle-to-grid gives the ground domain its own instance of the generalisation, which matters because it removes the objection that anchored collapse is a story about aircraft. At 7.4, 11 and 22 kW of contracted export — anchors up to 76.4 A — the anchored filter recorded zero unsafe episodes in 600 sessions each, while the null-input filter dropped the contracted load in **600 of 600** at every power.

7 Air: delivery rotorcraft and eVTOL

This is the domain the generalisation exists for, and the one where the null-input filter does not merely under-perform but loses the aircraft. On identical seeds, 600 flights each: the anchored filter recorded 0 unsafe-while-certified episodes; the null-input filter starved the rotors in **600 of 600**, serving a mean of 0.0 steps before brownout. It does not fly the aircraft for a while and then fail. It never flies it at all.

The two-sided admissible set is genuinely an interval: 20,000 flight states scanned densely returned 0 disconnected sets, with every bisected edge within one grid step of the dense scan. Across the whole flight envelope — payload × altitude, gusts to $\sigma = 0.5$ (3,000 flights), cold soak at 2 000 m, motor-out to +100 % power (3,000 flights) — nothing was breached while the certificate reported feasible.

7.1 The certificate refuses to fly the calibrated cell

The most interesting aerial result is a refusal. With the model's own calibrated coefficients — an energy cell — the anchored filter reports the interval **empty before takeoff** at and above 1.2C of hover draw. The admissible width collapses smoothly toward it: 5.06 A at 0.6C, 1.28 A at 1.0C, 0.16 A at 1.1C, empty at 1.2C (Fig. 2c).

This is not a filter failure. It is the filter working as a *design tool.* The correct engineering answer to "may I fly a rotorcraft on an energy cell" is no, and here that answer arrives before the aircraft leaves the ground rather than as a thermal event in flight. A runtime guard is not normally able to produce it, because a runtime guard is only ever asked about the vehicle it is already installed in.

The remaining aerial experiments therefore use a power cell — 0.40× resistance, 0.60× capacity — which is the largest single modelling liberty in this work and is flagged wherever it applies.

7.2 The reserve warns, and the takeoff transient sizes the pack

Across 1,200 flights, *every* closure preceded its breach: 0 breaches with no prior warning, median lead 24 s, 5th percentile 16 s, 95th percentile 62 s. Under gusts the result holds at every variance up to $\sigma = 0.5$ with 0 unsafe episodes in 3,000 flights — once the *live* closure is distinguished from a debounced one. Requiring three consecutive empty steps before declaring closure costs up to 96 s at the highest gust variance, and measuring lead from the debounced signal made an earlier version of this experiment report *negative* lead times for a filter that had already closed correctly.

A full delivery sortie — takeoff, climb, cruise, hover over the drop, return, land — shows the reserve tracking the mission without the filter being told a mission plan exists: 3.07 A in takeoff, 16.61 A in climb, 31.22 A in cruise, 26.23 A over the drop (Fig. 4).

Takeoff is the binding phase, and it is what sizes the pack. On 6S2P the certificate refuses a 1.75× takeoff transient outright; on 6S3P it allows it and 34.2 % of sorties complete; on 6S4P, **78.5 %**, with 0 unsafe episodes throughout. The frontier is actionable, which is what separates a design tool from a complaint.

Payload and density altitude close the envelope through the same physical term — momentum theory places both inside $T^{3/2}/\sqrt{\rho A}$ — and the reserve falls with each: $\rho = -0.981$ against payload and $\rho = -0.998$ against hover power, the shared term, and negative against altitude in *every* payload stratum. Pooled across the grid the altitude correlation is a non-significant -0.192 ($p = 0.359$), because payload spans a far larger range of hover power and swamps it; the within-stratum test is the one that answers the question.

An eVTOL air taxi hovering at 3.09C certifies with zero unsafe episodes, a median hover endurance of 3.0 minutes, and interval width predicting that endurance at $\rho = +0.694$.

7.3 Which edge binds, and why cold is the aerial problem

The two-sided certificate makes it possible to ask which constraint is actually limiting, and the answer moves with temperature in a way that is not intuitive. Table 2 reports the binding edge for a quadrotor at 2 000 m across an 70 K air-temperature range.

Two things follow. First, the thermal constraint — the one an aviation engineer would expect to dominate a high-power pack — essentially never binds; *voltage sag* does, everywhere below freezing. Second, the envelope closes outright in 43 % of states at -30 řC: a cold-soaked delivery drone at altitude is not marginally degraded, it is refused. Only at $+40$ řC does thermal appear at all, and even there it accounts for 3 of 400 states.

An operator planning winter service on the basis of thermal derating would be planning against

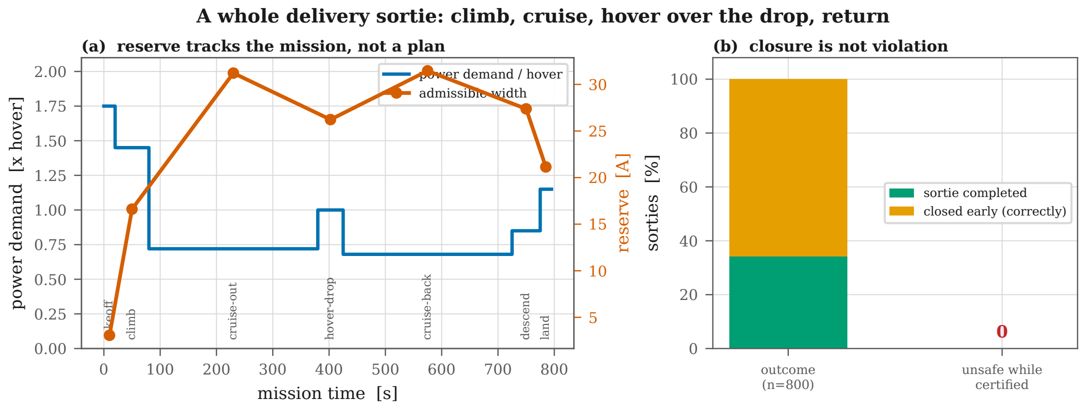

Figure 4: A whole delivery sortie. The reserve opens in cruise, where power demand is low and forced-air cooling is high, and closes over the drop. The filter is never told a mission plan exists; it sees a load that changes and re-derives both edges every step. Closure is not violation: the amber fraction is the certificate correctly refusing to continue.

Table 2: Binding edge for a delivery quadrotor at 2 000 m density altitude, 400 flights per node. Warm, the actuator ceiling limits and the aircraft is thrust-limited. Cold, series resistance climbs and *voltage sag* limits — and the interval closes entirely below −20 řC. Nothing was breached while certified at any node.

Air temp.	Mean reserve	Voltage	Thermal	Actuator	Closed
−30 řC	2.09 A	228	0	0	172
−20 řC	7.11 A	298	0	0	102
−10 řC	13.64 A	354	0	0	46
0 řC	20.20 A	279	0	120	1
+10 řC	25.30 A	108	0	292	0
+20 řC	27.35 A	45	0	355	0
+30 řC	28.01 A	9	0	391	0
+40 řC	17.73 A	0	3	251	146

the wrong constraint. The certificate identifies the right one without being told which to look for, because it evaluates all of them.

7.4 What forty sorties a day actually costs

A delivery pack lands at up to 44 °C, and the throttled margin puts the admissible ceiling at 32.5 °C — so the certificate refuses to charge it. Measuring how little charge a refused session delivers measures the refusal. The operational question is how long the pack must sit, and the filter answers it directly, because *the wait is the time until the interval opens.*

Of 4,000 turnarounds, 18 % could start at once; the rest waited a median of **60.8 minutes** (95th percentile 100.9), and none was still refused after two hours, with zero violations throughout. On a single pack that is 14.5 sorties/day. Swapping packs so the cool-downs overlap gives 37.2 sorties/day, and holding a 40/day target requires **2.8 packs in rotation**.

This is why delivery operators swap packs rather than wait. The contribution is that the certificate produces the number instead of the folklore.

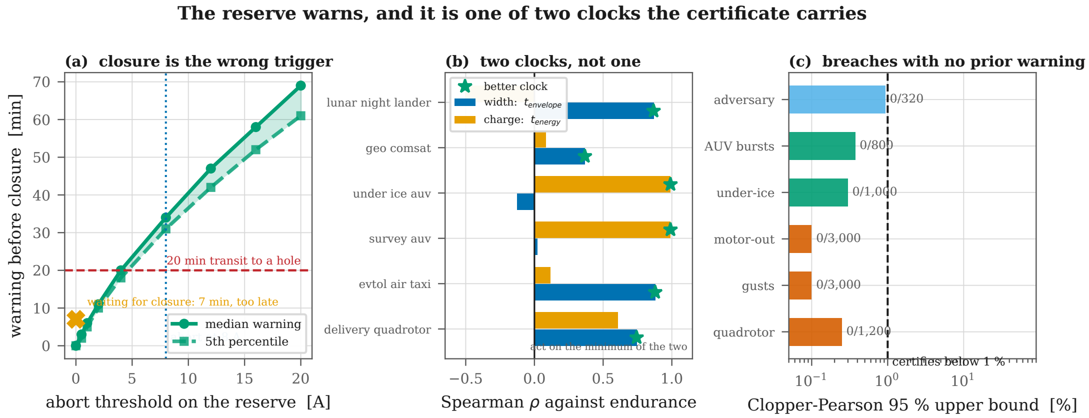

Figure 5: The reserve warns, and it is one of two clocks. (a) Under ice, closure itself leaves only 7 minutes before a breach against a 20-minute transit; aborting on a reserve threshold gives warning ahead of closure, and 8 A covers the transit. (b) Interval width predicts endurance where the envelope is what closes and does not where the pack simply empties; the certificate carries both clocks. (c) Breaches with no prior warning, across every stress: all zero.

7.5 An adversary

A cross-entropy adversary was given 320 independently drawn packs and 30,720 flight profiles, free to command anywhere inside the certified interval and optimising the demand and airspeed schedule to maximise peak temperature. It reached 79.2 % of the limit while the certificate was open and broke nothing: 0 violations, CP-95 upper bound 0.932 %, past the 299 cells needed to certify below 1 %. The statistical unit is the *cell*, not the sequence, because an earlier adversarial claim in this line of work rested on 40 cells and bounded nothing below 7.2 %.

8 Water: survey AUVs, gliders, under-ice vehicles

The hardest domain, for reasons that have nothing to do with the theorem. A pack inside a pressure hull is thermally isolated from an excellent heat sink — 0.0069 W/K per cell against a car's 0.0799, about one 12th — and the water is 2 °C, so both intuitions are wrong at once: it heats slowly *and* it is very cold.

It is also the cleanest separation of the anchored generalisation from anything about flight. A survey AUV sitting perfectly still on the seabed still cannot command zero: its computer, inertial navigation and acoustic modem draw 25 W regardless, an anchor of 0.60 A. Across idle, survey and transit the anchored filter recorded zero unsafe-while-certified episodes, and the null-input filter browned out the vehicle in **800 of 800 — including the case where nothing is moving.**

A sealed hull certifies: 3,000 dock recharges in 2 °C water, 0 violations. A six-month deployment with no recalibration: 547 recharges, zero violations. And 180 days with no ground truth: a fixed bound and a daily oracle record identical safety *and* identical delivered charge.

8.1 The constraint the theorem does not certify is the one that binds

This experiment was written to show the binding constraint *switching* from thermal to plating as the water cools. It never switches, because it was never thermal. **Plating binds in 100 % of states at every water temperature from** -1.8 **to 25 °C, and temperature binds in none** (Fig. 3b), across 4,200 trials. A sealed hull charges so slowly that the pack never approaches its thermal limit.

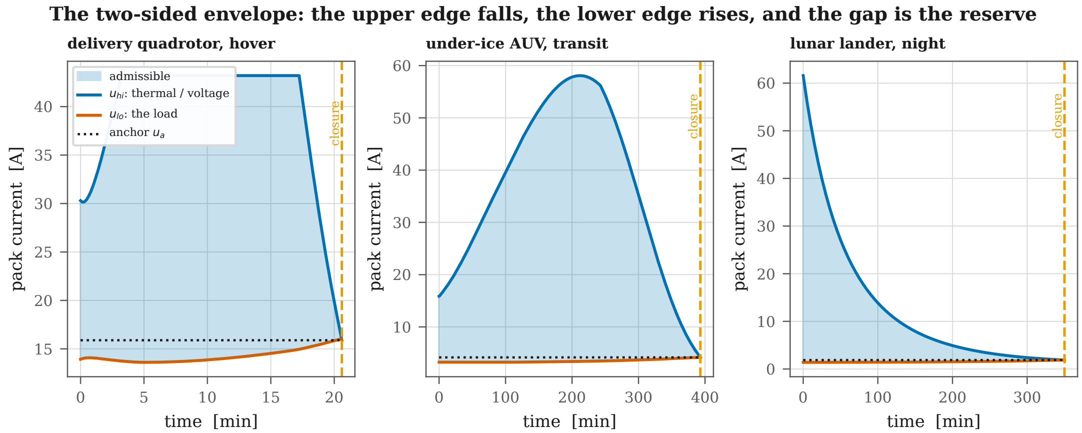

Figure 6: The two-sided envelope in three vehicles. The upper edge falls as the pack heats and sags; the lower edge rises as the load costs more current at lower terminal voltage; the gap between them is the reserve, and closure is where they meet. The dotted line is the anchor — the current the irreducible load demands, which is where the lower edge lands. On a charging cell that line sits at zero and the whole lower half of this picture is absent.

What changes with temperature is which plating mechanism binds: the temperature-dependent current cap takes 35 % of cases at 25 °C and 44 % at 2 °C.

That has a consequence worth stating plainly rather than dressing up. The certified set is $\{T, V\}$; plating is enforced and monitored but never certified, because $u = 0$ does not clear the plating margin at high state of charge and so no one-step condition can make it invariant. **On this vehicle, in this medium, the binding constraint is the one the theorem does not certify.**

Two downstream sweeps inherit it and both come back flat. Delivered charge does not move across ten years of dormancy (spread $2.0e-08$ SOC) or 6 000 m of depth ($7.6e-08$), because neither channel touches the plating cap. Those spreads are numerical noise and no rank correlation is reported over them — an earlier version did, and reported $\rho = +1.000$, $p = 0.0002$ for a flat line. *The certificate is not tolerant of dormancy and depth so much as blind to them.*

8.2 A negative result, and the design rule it forces

Under an ice shelf there is no abort-to-surface: a closed envelope is a lost vehicle. The question is therefore not whether the certificate is correct but whether it is *early*.

Waiting for the interval to close is too late. Closure is the moment the envelope is gone, so it offers no warning ahead of itself; what it does offer is the interval before the vehicle actually breaks something, and across 1,000 missions that is a median of 7 minutes. Against a 20-minute transit to a known hole in the ice, it covers the return in 0 % of episodes. 0 breaches occurred without a warning — the certificate is correct — and correct 7 minutes before the damage, when the vehicle needs 20, loses the vehicle anyway.

The fix is not a better certificate. It is to stop using closure as the trigger. The reserve is available at every step and falls smoothly toward zero, so the abort can fire on a *threshold* rather than on exhaustion (Table 3). Aborting at 8 A of remaining reserve gives 34 minutes of warning *before* closure at the median and 31 at the 5th percentile — covering the transit in every episode, with the 7-minute post-closure margin still in hand on top of it.

It is worth stating what this rule replaces, and being careful about it, because §11 tests the

Table 3: Under-ice abort rule. Two different quantities are involved and conflating them would overstate the result. Waiting for closure gives *no* warning before closure by definition — it *is* closure — and leaves only 7 minutes before an actual constraint breach. Aborting on a threshold in the reserve gives warning *ahead* of closure, which is the quantity a transit needs. Only the two-sided certificate has a reserve to threshold.

Abort trigger	Median warning before closure	5th percentile	Covers 20-min transit
wait for closure	0 min (by definition)	0 min	0 %
2 A	11 min	10 min	0 %
4 A	20 min	18 min	54 %
8 A	**34 min**	**31 min**	**100 %**
16 A	58 min	52 min	100 %

replacement and finds it does *not* dominate here. Current practice for autonomous underwater endurance is a fixed reserve fraction — turn back at 30 % state of charge, say — chosen conservatively because the quantity that matters, how much admissible current remains, is not observable. The natural claim is that such a constant must be simultaneously too cautious with a fresh warm pack and not cautious enough with a cold aged one.

Measured, that claim does not hold underwater. Across a population spanning the full parameter envelope, a well-chosen fixed threshold matches the reserve rule at equal safety (§11) — because an AUV's mission ends when the pack empties, and state of charge is a perfectly good predictor of that. The reserve threshold's advantage appears where the mission ends for some *other* reason, which underwater it does not.

8.3 Four decades of load, one certificate

A buoyancy glider drawing 0.5 W and an under-ice AUV drawing 175 W differ by a factor of 350 in load. Both produced single-interval admissible sets in every state scanned and zero unsafe episodes. What differs is the reserve *ratio* — 1575 for the glider against 7.7 for the AUV — which is the dimensionless statement of how much room each has relative to what it must draw.

9 Vacuum: LEO, GEO, deep space, Mars, lunar night

Space is where the certificate's least glamorous property becomes the decisive one. The filter requires no identification of the plant: no recursive least squares, no observer to converge, no ground station in the loop. On a car that is a convenience. On a spacecraft that has not been touched since integration, whose ground link is hours of light time away, and whose battery has taken 500 krad of ionising dose in the interim, **it is the only reason a fixed pre-launch parameter set is admissible at all.**

Space is also where the *form* of the guarantee matters as much as its content. Flight software is qualified by argument about worst-case behaviour, and an iterative solver is difficult to argue about: its run time depends on data that has not happened yet, its convergence depends on conditioning nobody can bound in advance, and its failure mode is to return late rather than to return wrong. A projection that always executes the same bounded sequence of arithmetic operations, allocates nothing, and branches a fixed number of times is a different kind of object to qualify. It is the difference between arguing that a component usually finishes and proving that it always does.

Table 4: Five mission classes and what each one makes hard. The certificate is unmodified across all of them; only the cooling law, the pack and the irreducible load change.

Mission class	**Defining stress**	**What it makes hard**	**Measured result**
LEO smallsat	5,662 cycles/year, no technician	count, not severity	28,307 charges, 0 violations
GEO comsat	72-min eclipse, payload cannot go quiet	depth of discharge at end of life	every duration completes, 0.0 % DoD
Deep-space cruise	4–250 K sink, hours of light time	recalibration is impossible	oracle buys 0.00 SOC points
Mars surface	80 K diurnal swing, 6 mbar CO_2	two cooling laws at once	2,400 trials, 0 violations
Lunar night	354 h at 100 K, heater is the load	both edges push on temperature	survival frontier measured, not passed

9.1 Four environments

Low Earth orbit is defined by its count rather than its severity: at a 92.9-minute period, 5,662 charge cycles a year. Five years — 28,307 charges, never recalibrated — produced 0 violations (CP-95 0.0106 %), with delivered charge on the final quarter matching the first to three decimal places.

Geostationary orbit is the opposite duty cycle: about ninety eclipses a year clustered into two seasons, the longest 72 minutes, with a payload that is not allowed to go quiet. Every duration out to the seasonal maximum completed, at a median depth of discharge of 0.0 %, with zero unsafe episodes.

Deep-space cruise radiates to a 4–250 K sink and nothing else: 3,000 draws, 0 violations, with the Stefan–Boltzmann cooling channel verified monotone in temperature numerically rather than assumed. **The Martian surface** adds 6 mbar of CO_2 to radiation across an 80 K diurnal swing: 2,400 trials at hourly resolution, 0 violations, with the sum of the two cooling terms confirmed monotone.

Ionising dose enters as one more monotone channel: zero violations to 500 krad, with delivered charge falling gently and monotonically in dose.

9.2 Why the count is the hard part in low Earth orbit

The LEO number is easy to misread as routine. Fifteen and a half charge cycles a day for five years is 28,307 cycles executed by a filter given its parameters before launch and sent nothing since. Over that span the pack's series resistance grows, its capacity fades and its plating margin tightens — and the certificate does not know any of it. It does not need to, because a monotone *bound* suffices where a point estimate would be required, and the bound was known at integration.

The observable consequence is that delivered charge on the last quarter of the mission matches the first to three decimal places. An operator's usual expectation is that battery performance is a slowly degrading resource requiring periodic re-planning. Here it is flat — not because the pack has not aged, but because the certificate was never relying on the pack being new.

9.3 Two inversions of terrestrial intuition

Losing the radiator helps. Sweeping εA down by two orders of magnitude produced 0 violations at every point — and delivered charge *rose*, from 0.736 to 0.759 (Fig. 3c). On a car, losing cooling costs charge. Radiating to a 4–250 K sink, a healthy radiator makes the pack *cold*, series resistance climbs, and the voltage limit arrives sooner; shrinking it lets the pack keep its own dissipation. In this domain the pack is cold-limited, not heat-limited, and a thermal-control failure of this kind is not an emergency.

Lunar night is a cold problem, so capacity is the wrong lever. A lander on 2.9 kWh survives about 5.3 hours of a 354-hour night, and quadrupling the pack buys only a couple more — because the envelope closes on cold-induced voltage sag, not on stored energy. Insulation is the lever: dropping emissivity from 0.72 to 0.05 takes survival from 5.3 to **54.3 hours**, at which point the pack stops being cold-limited and meets its energy ceiling of 61.3 hours. No configuration survives the full night, which is the honest answer and the reason real landers that survive lunar night carry radioisotope heaters rather than bigger batteries.

These two results share a structure worth naming. Terrestrial battery engineering is organised around removing heat, because on Earth the ambient sits close to the cell's operating temperature and the failure mode is thermal. In vacuum the ambient is 4 K and the failure mode inverts: the pack's problem is *retaining* heat, series resistance is the mechanism, and voltage sag is how it presents. A method tuned to terrestrial intuition — one that treats cooling as unambiguously good — would prescribe exactly the wrong design for both the deep-space probe and the lunar lander. The certificate makes no such assumption. It evaluates the constraints it is given under the cooling law it is given, and the inversion falls out.

9.4 No ground in the loop

Over 1,825 charges — five years — a fixed datasheet bound and an oracle recalibrated every single day record identical safety *and* identical delivered charge: the gap is 0.00 SOC points. In deep space, recalibration buys not just no safety but no performance either, because the binding constraint is the plating corner rather than resistance.

Solar-array degradation, swept to 60 years — three times any real mission, down to 29.8 % of beginning-of-life output — never becomes the binding constraint at all, and delivered charge varies by 0.0009 SOC across the whole sweep. The separation between what the vehicle can *request* and what the filter can *certify* is not marginal here.

9.5 The one fault the theorem cannot survive, measured anyway

Every other fault model in this work corrupts the plant or the sensor. A single-event upset corrupts the *certificate*: a heavy ion flips a bit in the state word the projection is about to read, so the filter computes an admissible interval for a spacecraft that does not exist and commands it to the one that does.

No one-step theorem survives corruption of its own input, and claiming otherwise would be dishonest. What can be measured is the shape of the exposure. Across 40,000 bit flips into state of charge, temperature and the RC overpotential: 5.80 % are caught by a three-comparison range check on the state word; 83.31 % are benign or leave the command *more* conservative; and 10.88 % produce a more aggressive command than the clean one. Of those, the worst overshoot actually realised in the plant was **0.000 °C** — none breached anything in one step. The deliverable is the range check and the number beside it, not a theorem.

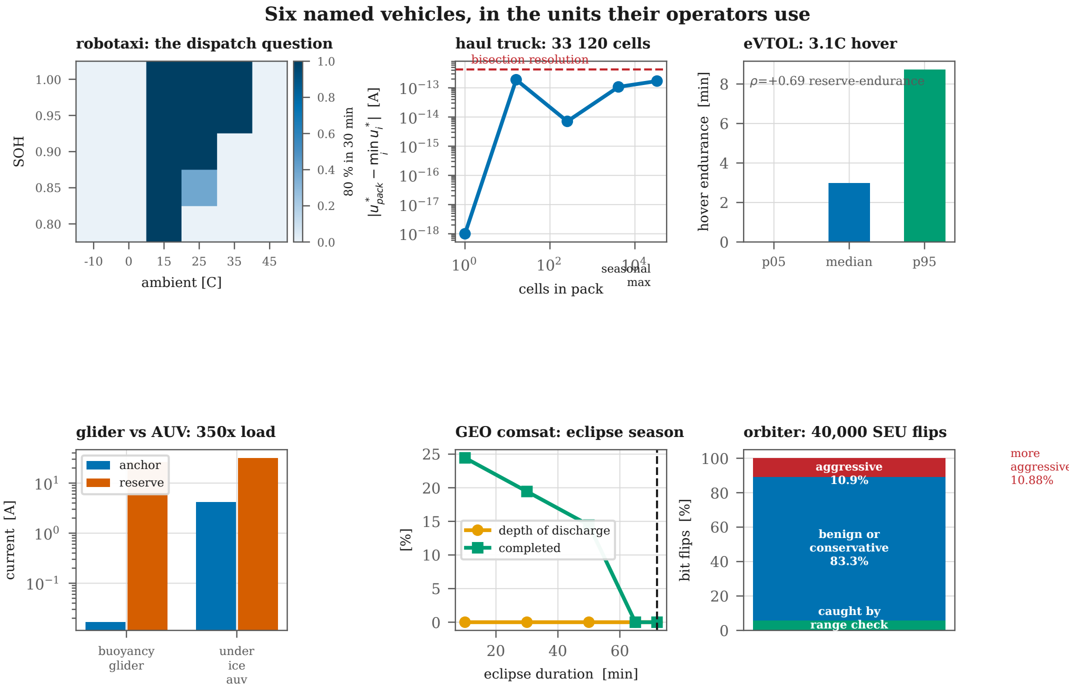

Figure 7: Six named vehicles, in the units their operators use: robotaxi dispatch feasibility over state of health and ambient; the weakest-cell result at 33,120 cells; eVTOL hover endurance; a glider and an AUV four decades of load apart; a GEO eclipse season; and single-event-upset containment on a high-radiation orbiter.

10 Compared to what?

Every safety number so far has the form "the filter violated nothing". That is necessary and not sufficient, because a controller that refuses to charge violates nothing either. The claims that carry weight are comparative, and until this section the only comparison was against a crippled version of the same method. Table 5 puts the certificate against what is actually deployed and what the literature actually recommends: 250 sessions per controller, identical plants, identical seeds, identical margins.

Against what ships today, carefully. A production pack is limited to a single conservative C-rate chosen offline and applied whatever the present state happens to be. It is tempting to pick such a rate, measure the gain against it, and quote the number — and that is exactly what an earlier version of this section did, reporting +16.4 points against 0.5C. **That number measured our own choice of baseline.** Sweeping the whole plausible range shows why: a fixed 1.0C is also safe on that population, and delivers *more* charge than the certificate, which carries a pessimistic estimator and a 12.5 K throttle that a well-chosen constant does not.

The comparison only becomes meaningful when the fixed rate is tuned and evaluated on *different* populations, which is what §10.1 does. A manufacturer does not get to choose a de-rate knowing the fleet it will meet.

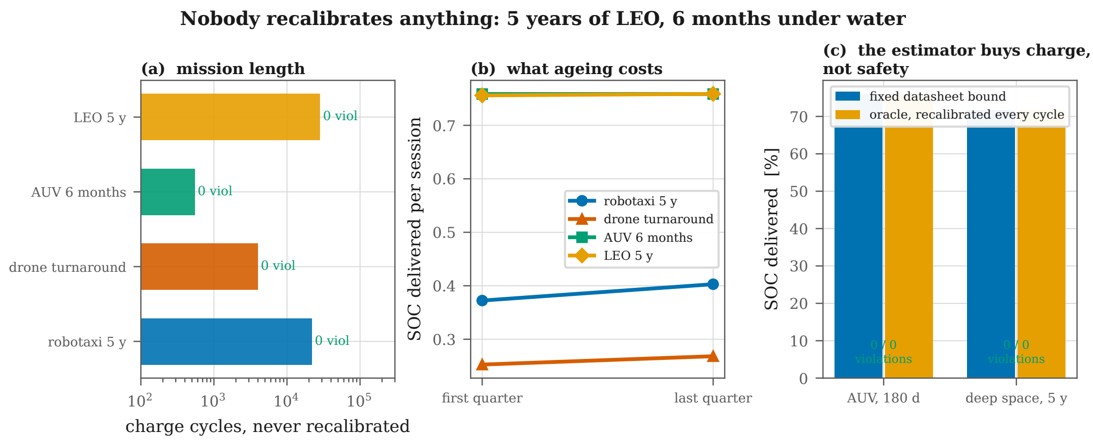

Figure 8: Nobody recalibrates anything. (a) Mission length by domain, on a log scale, with violations annotated. (b) What ageing costs across each mission: delivered charge per session, first quarter against last. (c) A fixed datasheet bound against an oracle told the plant's true resistance before every cycle — identical safety, and on these two missions identical delivered charge.

Table 5: Four controllers on the same plants. Cost is counted in one-step *model evaluations* rather than seconds, because wall-clock depends on the language and the machine while the evaluation count is a property of the algorithm — and is what a task budget is sized around.

Controller	Violations	Mean SOC	Evals (med)	Evals (max)	Worst case kno
CC-CV at 0.5C (safe on this population)	0	0.477	1	1	—
CC-CV at 1.0C (also safe here)	0	0.749	1	1	—
CC-CV at 1.5C (first unsafe rate)	—	—	1	1	—
MPC, horizon 1 (SLSQP)	0	0.634	11	644	no
MPC, horizon 3 (SLSQP)	0	0.642	94	502	no
ZEROGUARD	**0**	**0.641**	20	**20**	**yes,** $=$ **20**

10.1 A de-rate, tuned honestly, then deployed

The fair experiment tunes the constant on evidence a manufacturer could actually have, then ships it into the world.

Stage 1, characterisation. On a benign fleet — 25 °C ambient, cells not yet aged — the fastest fixed rate with zero violations is **1.2C**. That is the best de-rate this evidence supports, and a manufacturer choosing it would be doing nothing wrong.

Stage 2, deployment. On the fleet a vehicle actually meets — ambient -10 to 40 řC, cells out to end of life — that same 1.2C rate violates in **26.2 % of 400 sessions**. The certificate, given the same widened population and re-tuned not at all because it has nothing to tune, violates in **0** (CP-95 upper bound 0.746 %).

What safety costs a constant. To be safe across the deployment fleet, a fixed rate must fall to 0.5C. Against that correctly conservative constant the certificate delivers +1.0 points (+2.0 %) — a real gain, and a modest one.

So the argument for the method is not that it charges dramatically faster. It is that **no constant is simultaneously safe across the deployment fleet and fast on the benign part of it**, because a constant cannot see which fleet it is in. The certificate can, at 123.9 μs and without identifying anything. Charge parity-or-better is the by-product; the transfer of the guarantee is the result.

Table 6: Mission delivered (median, minutes) at two recovery targets, 300 missions per domain. "–" means no threshold in that family reaches the target. The certificate's reserve wins where the envelope is what closes and loses where the pack simply empties — which is the two-clock result of §21 appearing again as an operational number.

Domain	**Recovery target**	**Fixed SOC**	**Reserve**	**Both**
Delivery quadrotor	80 %	6.2	**8.0**	8.0
	95 %	–	–	–
Under-ice AUV	80 %	**201.5**	193.5	201.5
	95 %	**201.5**	193.5	201.5
GEO comsat	80 %	**62.5**	31.0	56.5
	95 %	**62.5**	8.0	56.5

Against MPC: the same problem, solved two ways. This is the comparison that matters for the run-time claim, because that claim is only interesting if the solver is solving the same problem. It is: horizon-1 MPC and the bisection agree on delivered charge to **0.66 SOC points**. The certificate is not a cheaper approximation of MPC — on this problem it is an exact solver for it.

The difference is what it costs to be sure. The bisection spends at most 20 model evaluations, and that is a *bound*: proved in §4.3, known before any data arrives, never exceeded. SLSQP's worst observed cost was 644 evaluations, a factor of 32.2 higher — and that number is not a bound. It is whatever the iteration cap happened to allow, its iteration count varied by 59 across states, and removing the cap removes the number. A safety task slot can be sized around the first quantity and cannot be sized around the second.

What this does not show. SLSQP is a general-purpose nonlinear solver and a specialised QP or an explicit-MPC lookup would be faster; the comparison is against the standard approach as it is usually implemented, not against the best conceivable optimiser. What no optimiser removes is the data-dependence: the argument here is about the *shape* of the cost, not its constant.

11 Against each domain's own stopping rule

§10 compared the certificate against CC-CV and MPC on a charging pack. That is the right comparison for the ground domain and the wrong one for every other, because the decision a flying, swimming or orbiting vehicle makes is not *how much current* but *when to stop*. Each domain has an incumbent rule and each is a fixed fraction: land at 20–30 % state of charge, turn back at 30 %, do not discharge past 50–60 % depth in GEO.

Comparing against them requires care about what "safer" means. It is not "no violation" — a rule that stops immediately violates nothing. It is *did the vehicle get home*. So every rule is scored on two axes: mission delivered, and the fraction of episodes that retained enough envelope to complete a stated recovery manoeuvre after stopping. Rules are then compared **at matched safety**, which is the only fair way to compare points on a trade-off curve. All rules are evaluated on the *same* trajectories, so the comparison carries no sampling noise.

Where it wins. On the rotorcraft the certificate's reserve delivers +28 % more mission than the best fixed SOC threshold at an 80 % recovery target. The reason is the one §7 already identified: a quadrotor's envelope closes on voltage sag, not on charge, so state of charge is simply not the quantity that is running out. A fixed SOC reserve does not fire before closure at all — it is not conservative, it is *uninformative*.

Where it does not. Underwater and in GEO eclipse, a well-chosen fixed SOC threshold matches or beats the reserve rule at every safety target tested. Those missions end when the pack empties, and state of charge predicts that directly and cheaply. **This contradicts a claim we made before measuring it** — that a fixed reserve must be badly miscalibrated across a heterogeneous population — and §8 has been corrected accordingly. Across the full parameter envelope a constant threshold does fine, because the thing it is predicting really is close to a constant fraction of charge.

What to actually do. Neither clock dominates, which is precisely the prediction of §21: the width predicts time-to-closure where the envelope closes, the charge clock where the pack empties. Acting on the *minimum* of the two is never far from the best single rule in any domain, and is the only rule that works in all three — because it degrades to whichever clock is informative. That is the recommendation, and it is now a measured one rather than an argued one.

Above 90 % recovery on the rotorcraft, nothing works. No rule in any family reaches a 90 % target. The quadrotor's interval closes abruptly enough that no threshold on either clock buys reliable warning, which is a limitation of the vehicle rather than of the rule, and is reported rather than hidden by choosing a friendlier target.

12 The same standard in every domain

Two things were wrong with the per-domain comparisons, and both were ours.

The standard was applied unevenly. §10.1 caught a methodological error — a fixed de-rate tuned and evaluated on the same population — and corrected it for the *ground* domain by tuning on a fleet a manufacturer could characterise in advance and deploying on the fleet a vehicle actually meets. §11 then compared against the air, water and space incumbents *without* that correction: every fixed state-of-charge threshold there was swept against the very population it was scored on. The ground baseline was held to a stricter standard than the other three, which is backwards, and water and GEO — where the fixed rule won — are precisely the results the correction would move.

And the ground metric was the wrong quantity. §10.1 reports mean state of charge after a fixed horizon. No operator buys state of charge. A robotaxi that is charging is a robotaxi not earning, so the quantity is minutes until it can be dispatched again.

12.1 Tune where you can characterise; deploy where the vehicle goes

Each domain's incumbent threshold is chosen on a benign population — mild ambient, cells not yet aged — taking the most permissive threshold that still recovers at the stated rate. It is then deployed, unchanged, on the population the vehicle meets. The certificate reserve is put through the identical procedure and, as everywhere else, is re-tuned not at all because it has nothing to tune. Targets differ by domain and are stated, because in the aerial case no rule of *either* family recovers in 95 % of episodes and fixing the target there yields no comparison rather than a result.

The correction changes one domain and not the others (Table 7). In the air the tuned threshold **fails**: chosen on evidence a manufacturer could actually have, it recovers 86.0 % in characterisation and 76.0 % in deployment, below the target it was chosen to meet, while the certificate reserve goes

Table 7: Recovery rate in characterisation → deployment. 300 missions per population per domain.

domain	rule	char.	depl.	verdict
Air (≥80 %)	soc≤40%	86.0	76.0	**fails**
	certificate	86.0	85.7	holds
Water (≥95 %)	soc≤20%	100.0	100.0	holds
	certificate	100.0	100.0	holds
Space (≥95 %)	soc≤45%	100.0	98.3	holds
	certificate	97.0	96.7	holds

86.0 % to 85.7 %. That is the ground transfer result of §10.1 reproduced in a second medium and in a different currency — there it was violations, here it is failures to get home.

Underwater and in GEO the incumbent *survives* its transfer test, and the certificate remains behind on mission delivered: -4.5 % and -84.7 %. We looked for the methodological error that would overturn those two results, applied the correction that had overturned the ground one, and it did not overturn them. §21 already says why, and says it in advance rather than afterwards: a constraint-based clock can only predict endurance where closure is envelope-driven, and underwater and in eclipse it is energy-driven. The certificate is measuring the wrong thing there, and no amount of fairer baselining will change that.

12.2 Minutes back on the road

Scored on dispatch time instead of state of charge, over 1,200 sessions across the deployment fleet against the CC–CV rate §10.1 showed is fleet-safe, the ground picture separates into two facts the average was hiding.

The first is that the advantage is much larger than a mean SOC suggests. Within the 40-minute window, 45 % of sessions reach a 60 % charge under the certificate against 3 % under the fixed rate — 17.4× as many vehicles returned to service — and at 70 % the fixed rate reaches it in 0 % of sessions. A fleet is sized on how many vehicles come back, not on the mean charge of those that do.

The second is that the fleet mean of +1.2 points is an average taken across a change of sign, and is therefore not a fact about any vehicle. By ambient the certificate delivers +23.8 points at 15 °C and -29.7 at 40 °C, crossing over between 25 and 35 °C.

The reversal is not the theorem failing, and it is not a mystery. The thermal margin puts the certificate's effective ceiling at 32.5 °C; above that ambient the cell can never sit below it, so the filter refuses to charge. It is right to refuse — it breached nothing in 0 of 1,200 sessions — and refusing has a price. This is the passive-charging ceiling of §6 reappearing as a performance number rather than a safety one, and it is the same margin-sizing question §19.1 answers for pulses and does not answer here, because a charging session really does last half an hour.

13 Against the CBF quadratic program

§2.1 places this work among predictive safety filters and control barrier functions, and §10 then benchmarks it against CC–CV, a de-rate sweep and MPC. That leaves the obvious comparison unmeasured. The CBF quadratic program is *the* standard safety filter, and this method claims to replace the optimisation inside it, so the comparison is not optional.

Table 8: Exact plant, zero margin. 600 sessions per row; only the approximation is left.

filter	breaches	SOC
CBF-QP, $\gamma = 0.05$	0/600	0.208
CBF-QP, $\gamma = 0.1$	0/600	0.244
CBF-QP, $\gamma = 0.2$	0/600	0.319
CBF-QP, $\gamma = 0.4$	0/600	0.474
CBF-QP, $\gamma = 0.7$	0/600	0.697
CBF-QP, $\gamma = 1.0$	**42/600**	0.780
ZEROGUARD	**0/600**	0.782

The discrete-time CBF condition [23] asks, for each constraint written $h(x) \geq 0$,

$$h(x_{k+1}(u)) \;\geq\; (1-\gamma)\, h(x_k), \qquad \gamma \in (0, 1], \tag{5}$$

and the filter projects the request onto the resulting set. For a cell model $h(x_{k+1})$ has no closed form, so in practice the constraint is **linearised** in the input, $h(x_k) + (\partial h/\partial u)\, u \geq (1-\gamma)h(x_k)$, which makes the problem a QP. That linearisation is the entire difference between the two methods, and everything below is arranged to isolate it. Both filters are given the identical constraint set, margins folded in the same way, on identical plants and seeds.

The exact condition is this method. Take the CBF condition and enforce it on the *true* model instead of a linearisation. Because each h is monotone in u (§3), $\{u : h(x_{k+1}(u)) \geq (1-\gamma)h(x_k)\}$ is an interval; γ enters as an additive shift of the margin, and bisection finds the edges. Measured over 600 states with γ drawn from the sweep, the admissible set was a single interval in *all* of them (0 disconnected), and the bisection edges agree with a dense scan to 0.902 A against a scan step of 1.353 A.

So the contribution is not a rival to CBF. It is a statement of *when the CBF-QP can be solved exactly at fixed cost* — and for monotone constraints in a scalar input, the answer is always. That is a stronger position than beating the QP, and it is the one the geometry actually supports.

Where the linearisation shows. Run as deployed — pessimistic estimator, the 12.5 K margin — the CBF-QP breaches nothing at any γ in the sweep. It would be easy, and wrong, to read that as the linearisation being harmless. Between the worst-corner estimator and the margin there is enough slack to swallow a linearisation error many times the measured 0.68 barrier units; shrinking the margin does not isolate it either, because the pessimism alone suffices — tested to *zero* margin, neither filter broke. What the QP costs there is charge: at the best safe $\gamma = 0.7$ it delivers 0.572 against the certificate's 0.635, a gap of 6.4 SOC points.

To see the approximation itself, every other source of conservatism has to go: both filters are given the *exact* plant — no envelope pessimism — and no margin at all. A filter that evaluates the true one-step model is then exactly safe by construction, and anything that breaks is breaking on its own approximation. γ is swept rather than fixed, because $\gamma < 1$ holds back a fraction of the barrier and that reserve is precisely what would conceal the linearisation. $\gamma = 1$ is the matched condition: the CBF constraint is then $h(x_{k+1}) \geq 0$, which is the invariance condition the certificate enforces, and the only remaining difference is that one filter linearises it.

At the matched condition the linearised filter breaches in 7.0 % of sessions (CP95 8.96 %) and the bisection in 0 of 600, on the same plants, with a perfect model and no margin (Table 8). The

QP is not unsafe because CBF is unsound — the condition is the same one — but because it solves a linearised version of it.

It can of course be made safe by lowering γ, and that is exactly the trade: safe at $\gamma \leq 0.7$, where it delivers 8.5 SOC points less than the exact filter. The reserve γ holds back is charge the vehicle does not get, spent covering an approximation that did not have to be made.

What this does not claim. The QP is cheaper — 2 model evaluations against 20— and for a scalar input its solution is a clip rather than an iterative solve, so the run-time argument of §10 does not apply to it. Nothing here is a speed claim. The claim is that at equal safety the exact filter delivers more charge, and that for this constraint class exactness is available at a cost that is fixed and known in advance.

13.1 The firmware, compiled

Everything above is an estimate. Bytes of state were counted from a Python dictionary, table sizes from an interpolation error study, and "needs no FPU" from the observation that a midpoint is an integer operation. Those are reasonable ways to size a footprint and no way at all to establish one: *fits in a kilobyte* is a claim about a binary, and there was no binary.

`firmware/zeroguard.c` is one. The charge-direction certificate in C99, with the integer bisection and a float32 model, no heap, no recursion, no variable-length structure, and every loop bound a compile-time constant — so worst-case execution time is a property of the source rather than of the data, which is the property a safety task slot is actually sized against. Its constants are *generated* from the same calibration files the Python filter loads; had they been transcribed, the two could drift and every agreement number below would be measuring a copy of itself.

Built at `-Os` with `-Wall -Wextra -Werror` and no warnings, it occupies **2,064 B** of code and constants — 512 of that the open-circuit-voltage table — with 0 B of writable data and 68 B of RAM live across a call. One call is exactly 18 model evaluations.

Does it agree with the reference? Not bit-for-bit, and it was never going to: the reference computes in float64 and bisects in floating point, the firmware computes in float32 and bisects in integers. The question is which way the disagreement points. Over 20,000 states with an identical pack and identical margins, the two never disagreed about *feasibility* (0 mismatches), and the returned current differed by at most 5.5 mA against a bisection quantum of 8.2 mA — the firmware exceeded the reference by more than one quantum in 0 states. Where it was higher at all, the extra current is worth 0.013 mV of terminal voltage (0.04 % of the voltage margin) and 0.0000 K. The guarantee survives the port with the margin it already carries, and does not need a new one.

The build is for `arm64-apple-darwin25.6.0`, not a Cortex-M part, because no cross-compiler was available here; the byte counts for state, tables and constants are architecture-independent and the code size is not, so that number should be read as indicative for a different instruction set rather than as a measurement of one.

14 A policy through the filter

The position stated in §2 — the policy may be learned, the safety must not be — has so far been an argument rather than a result. Every experiment to this point either runs the certificate alone or sets it against a rival controller. None puts a policy *through* it, which is the arrangement the method is for. A safety filter is not a controller; it sits between an arbitrary controller and the plant,

Table 9: Transparency as a curve. 80 sessions per row.

requested	filter intervenes	unfiltered breaches
0.3C	0.0 %	0/80
0.5C	0.7 %	0/80
0.8C	37.9 %	0/80
1.0C	76.3 %	3/80
1.5C	93.8 %	15/80
2.0C	98.2 %	60/80
3.0C	100.0 %	77/80

and two properties decide whether it has done its job. They pull in opposite directions, which is why both have to be measured on the same runs.

Containment. An unsafe policy, filtered, must become safe — not safer.

Transparency. A policy that was already safe must be left alone. A filter that clips a good controller is buying safety with performance that did not need to be spent, and the standard way to conceal that is to test only aggressive policies.

Four policies, one plant. Each is run unfiltered and filtered on identical plants and identical seeds over 400 sessions: a greedy controller that always demands $u_{\max}$; a naive thermal heuristic that runs at full current until the cell feels warm and then tapers, which is the shape of controller a competent engineer writes without a model; a *learned* policy; and production CC–CV at 0.5C, included because transparency needs a policy that is genuinely safe to begin with.

The learned policy is a smooth state feedback $u = u_{\max}\,\sigma(w^\top[1, \mathrm{soc}, T, V])$ whose four weights are fitted by cross-entropy search to maximise delivered charge on a *nominal* cell, with no safety term of any kind in the objective. That is the honest version of the experiment: not a policy trained to misbehave, but one trained to be greedy, fitted against the cell a practitioner would have on the bench, and given no reason to care.

What the search returned is worth reporting rather than dressing up. It converged on the boundary — a bias of +7.6 saturates the sigmoid, so the fitted policy is greedy in all but name, and its unfiltered behaviour is indistinguishable from the greedy controller's. That is not a failure of the search. It is what maximising delivered charge with no safety term *is*, and it is the reason a learned controller cannot be asked to limit itself.

Containment. 3 of the 4 policies breach a certified constraint unfiltered, the worst of them in 100 % of sessions. Through the filter, all of them breach nothing: the learned policy goes from 400/400 to 0/400.

Transparency. On the policy that was already safe the filter intervenes in 0.39 % of steps and costs 0.003 SOC points. That single number is weak evidence on its own, because transparency is not a property of one policy, so the requested C-rate is swept from below the admissible ceiling to far above it (Table 9). Intervention rises monotonically with the request; the filter is untouched up to 0.5C and the unfiltered policy is unsafe from 1.0C. The order of those two matters: clipping begins *before* breaching does, which is what a filter that is neither indifferent to the request nor asleep looks like.

And it is worth doing. The obvious objection is that containment is trivial if the filter simply refuses everything. It does not: the filtered aggressive policy delivers +16.7 SOC points more than the safe CC–CV protocol it replaces, with the same zero breaches. The comparison is deliberately against that protocol and not against the unfiltered run, because the charge the unfiltered run delivered was bought with violations and is not a baseline anyone may claim against.

15 The weakest cell, in motion

§6 established the pack lemma statically: for a 33,120-cell truck pack the certified pack current equals the weakest cell's to 10^{-13}, at one state. That is a snapshot. It says the limit is set by whichever cell is worst *at the instant of evaluation*, and says nothing about what happens once the cells stop sharing a state — which they do immediately, because a pack is not thermally uniform and its cells are not identical.

The experiment runs in two phases, and the division of labour is the point. First the pack of 128 individually simulated cells — each with its own resistance, capacity, plating scale and cooling coefficient — drives 2 laps of the EPA schedules of §19. Nothing is filtered and nothing is claimed there; the cycle's job is to *produce* a diverged pack from an external input rather than from a spread we invented, and it is also how a real pack arrives at a charger. Then the diverged pack is fast-charged, which is where the certificate binds — in 100 % of steps — and where the pack claim carries weight.

An earlier version of this experiment tried to test the pack rule on the regenerative braking *within* the cycle and had to be abandoned. Instrumented, the certificate bound in about 1 % of braking steps on US06 and in none at all on UDDS or HWFET: for this pack and these schedules, regeneration is simply not constraint-limited at moderate SOC, so every rule — including a deliberately wrong one — returned a clean sheet. That is a fact about the duty cycle, not a result about heterogeneity, and reporting the resulting agreement would have meant nothing.

The binding cell moves, and it is the wrong cell. Driving leaves 1.1 K and 1.3 SOC points of spread. Over the charge that follows, as many as 9 distinct cells take turns being the binding one, handing over in 5 % of steps, so an implementation that identifies the worst cell once and caches it is unsound; the minimum has to be taken every step. It is the worst-resistance cell in only 0 of 24 packs.

More interesting is *which* cell binds. Its mean temperature rank is 0.29 against 0.50 for an indifferent draw, and it is the coldest cell in the pack outright 7 % of the time. This is the reverse of the intuition a lumped thermal model encodes, and it follows directly from §3: in the charge direction the plating cap tightens as a cell cools, so a cold cell is a weak cell. The hot-cell heuristic is looking at the wrong end of the distribution.

What the lumped model breaks is not the theorem. The obvious shortcut is to evaluate the certificate once, on mean parameters at the mean state. That is what a pack-level model *is*, and it is what every simulation reporting a single pack temperature implicitly certifies against. It is given the same pessimistic resistance the real estimator carries, so the only thing it gives up is the spread.

It breaks no certified constraint: 0 of 24 packs, exactly like the minimum rule. That is the theorem doing its job and it is not the interesting line. What it breaks is plating — 596 of 3,072 cells (19.4 %), in 24 of 24 packs, against 0 for the minimum over cells — bought for +2.87 SOC points. Plating is enforced and never certified (§4), so no theorem is violated; the cells are damaged all the same, which is the entire reason the channel is enforced. A study that counted only certified channels would have reported the shortcut as safe.

Nor does this depend on an unusually loose pack. Sweeping the thermal spread from 1.1 K — tighter than any production specification — to 11.3 K, the lumped model plates cells at *every* point, worsening from 205 cells to 357, while the minimum over cells plates 0 anywhere in the sweep (1,562 against 0 in total, over 48 packs). The minimum is a requirement, not a preference.

16 External validation: the assumptions against measured cells

Everything to this point is simulation. The filter is checked against a reduced-order model, and that model was calibrated against a Doyle–Fuller–Newman simulation. That is a defensible chain and it is still a chain of models, so the fair question is what happens when the assumptions meet a real cell.

This section answers it with the NASA Ames Prognostics Center of Excellence battery data set: 33 18650 cells cycled to failure with periodic electrochemical impedance spectroscopy, published by a different laboratory, years before this work, for a different purpose. Nothing here was produced by us.

The design point is that **the load-bearing assumptions are testable without any model at all.** The method does not need the ROM to be right about NASA's cells. It needs three things to be true of real cells, and each is a chemistry-independent statement about the shape of degradation.

16.1 The resistance bound holds

The filter never identifies resistance; it assumes a *bound* on it, taken from a datasheet before the vehicle ships. If real cells leave that bound, every certificate in this paper is void. Across 1,940 impedance measurements, **99.95 % lie inside the assumed bound** $s_R = 1.8$ (Fig. 9a). Restricted to the life the certificate actually claims — cells at or above 80 % of initial capacity, the standard automotive end of life — the figure is **100.00 % of 1,255 measurements**.

Over the full measured life, including cells NASA ran down to 0.033 of initial capacity, 1 measurement exceeds the bound, on cell B0043, by 0.1 %. That is reported rather than trimmed: the bound is tight, and it is not a comfortable over-estimate.

16.2 Heat generation is monotone in current, measured

Hypothesis (A2) is the other load-bearing assumption, and it too can be checked without a model — though the obvious way of doing it does not work, and why is worth recording. Correlating instantaneous $\mathrm{d}T/\mathrm{d}t$ against instantaneous current returns nothing usable: stratifying by $T - T_{\mathrm{amb}}$ to control for cooling gives rank correlations that *flip sign* between strata (-0.34 in the coldest, $+0.54$ in the warmest). NASA measures surface temperature, the core leads it by minutes, and over a charge whose total rise is under 3 K the thermal lag dominates the signal. Reporting either sign from that test would be an artefact of the instrument.

The lag-free formulation is an energy balance over a whole cycle, which does not care how the heat arrives: total ohmic generation goes as $\int I^2\,\mathrm{d}t$, so a cycle's peak temperature rise should increase with it. Across 2,764 measured discharge cycles, with currents to 4.04 A and temperature rises to 37.7 K, the rank correlation is $\rho = +0.714$ ($p < 0.0005$; Fig. 9b). The charge cycles give $+0.278$, which is weaker for a mundane reason: NASA charges almost everything at a constant 1.5 A, so there is little current variation to correlate against.

16.3 Where the model does not transfer, and whether that matters

The harder question is how the ROM itself behaves on a cell it was never fitted to — a 5 Ah NMC model applied to ∼2 Ah LCO cells with capacity rescaled and nothing else refitted. Measured one step ahead the question is meaningless, because NASA samples every ∼9 s over which the temperature moves about 0.01 K and predicting "unchanged" scores well. Integrated forward over a 300 s horizon under the measured current, the transferred model scores RMSE 0.842 °C against a persistence baseline of 1.096 °C — a skill of +0.231.

The safety-relevant question is not the magnitude but the *sign*. A model that over-predicts temperature yields a conservative certificate; one that under-predicts yields an unsafe one. **This transfer is not conservative**: the bias is -0.268 °C and it over-predicts in only 18.4 % of forecasts. What rescues it is the margin. The worst under-prediction at the 5th percentile is 1.11 °C against a thermal margin of 12.5 °C — a factor of **11** of cover (Fig. 9c). The margin was not designed for cross-chemistry transfer; it happens to absorb it, and that is a fact about this data rather than a guarantee.

16.4 The filter permits a real protocol, and refuses where it said it would

A safety filter that refuses everything is safe and useless. Given the end-of-life resistance bound and the full throttled margin — its most conservative configuration — the certificate permits **71.6 %** of NASA's measured charge steps, with a median headroom of 4.83× the applied current. Of the refusals, 46 % are cells measured above the throttled ceiling of 32.5 °C: the same frontier §6 predicts from the margin alone, appearing unprompted in someone else's measurements.

16.5 One test that did not work

It was tempting to go further and ask whether the cells the filter would have refused are the cells that actually degraded. The mechanism would have been clean — NASA cycled some cells at 4 °C, the plating cap falls to a 0.70C floor below about 5.5 °C, and plating is what degrades a cold-charged cell.

The association is not stable. Its sign *changes* with how the capacity baseline is defined — first measured discharge against the median of the largest few — because NASA's first discharge on 12 of 33 cells is partial, giving baselines as low as 0.068 Ah for a 2 Ah cell. Across the four defensible combinations of baseline and fade metric, ρ ranges from -0.358 to $+0.395$, with significant results of both signs, and ambient temperature and cycle count sit behind both quantities. All four are reported in Table-form in the results file rather than one. **A result that flips sign with a defensible change of definition is not a result**, and selecting the flattering version would be the most misleading thing this study could do.

17 Against a Doyle–Fuller–Newman plant

Every vehicle number in §6–§9 comes from a plant we wrote. The reduced-order model is defensible and it is still ours, so the question a battery reader asks is not whether this was tested on hardware but the sharper one underneath: *how do you know the ROM is right?*

DFN [8; 9**?**] is the field's reference model — porous-electrode transport, solid-phase diffusion, Butler–Volmer kinetics, parameterised and validated against cells by others — and it is a *different model class*, not the same equations with different numbers. That is what makes it worth running. The envelope of §5 varies resistance, capacity and plating scale *within* the ROM's structure; DFN

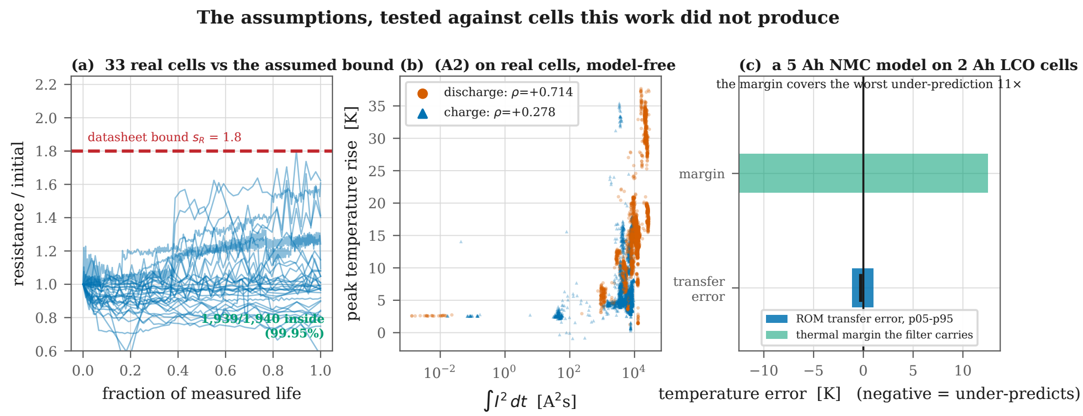

Figure 9: The assumptions against cells this work did not produce. **(a)** Measured resistance growth for 33 NASA cells against the datasheet bound the filter is given. **(b)** Hypothesis (A2), model-free: a cycle's peak temperature rise against its total ohmic generation. **(c)** The transferred model's error against the thermal margin that has to absorb it.

inflicts misspecification the envelope was never written to cover: concentration gradients the ROM has no state for, kinetics it approximates, and a plating potential that is an actual electrode quantity rather than an affine proxy. The filter sees only measured SOC and temperature, propagating its own relaxation state between them, because that is all a battery management system has.

What this is and is not. The ROM's coefficients were fitted to the LG M50, and the PyBaMM parameter set used here (OKane2022) describes that cell, so this is **not** an independent-cell test and must not be read as one. It is a model-class test: same cell, higher fidelity. §16 covers the other axis — 33 cells from another laboratory that nothing was ever fitted to. Neither substitutes for the other; together they bracket the two ways a model can be wrong.

Open loop, and the sign is the whole answer. Driving both models with the same current, the ROM disagrees with DFN on terminal voltage by up to 132 mV — 4.4× the voltage margin. Read as a magnitude that is alarming, and read as a magnitude it is also meaningless, because *the error is almost entirely in the conservative direction*: the ROM reads *above* DFN, so the filter throttles earlier than it needed to. For a cap, only the ROM reading *below* the truth endangers anything.

Separating the two changes the conclusion. Optimism reaches 137 % of the voltage margin somewhere on the trajectory — but that worst case is at 0.3C, where DFN peaks at 3.74 V against a 4.20 V limit, so it is arithmetic rather than risk. Restricted to steps where DFN is actually near a limit, the ROM is optimistic by 0 % of the voltage margin and 1 % of the thermal margin. Inside both. An earlier version of this experiment reported |error| and made 4.4×-margin *conservatism* look like a 4.4×-margin safety gap, which is the opposite of what the data says.

Closed loop. With the certificate computed on the ROM and the plant a full DFN cell, across 40 charging sessions: 0 voltage breaches, 0 temperature breaches. DFN's own peak values were 4.052 V against 4.20 and 32.5 °C against 45, so the cell was worked, not idled.

The plating result is the one worth pausing on. The ROM enforces an affine proxy; DFN reports the negative electrode surface potential difference, whose crossing of zero *is* the thermodynamic onset of lithium deposition. In 40 of 40 sessions the real electrode potential stayed above onset, with a minimum of +58.5 mV. The proxy the paper has enforced throughout, and has been careful never to certify, does the physical job it was built for — measured against electrode physics rather than against itself.

How wrong the physics can be. Scaling DFN transport properties away from nominal misspecifies the plant structurally, in ways no scale factor in the ROM can represent. The certificate holds with electrolyte conductivity at 20× below nominal and gives way at 50×; with negative particle diffusivity it held to the end of the sweep, 100× below nominal, so that figure is reported as where the sweep stopped and not as a measured floor.

This sweep was wrong the first time and the failure is worth recording, because it produced a clean result rather than an error. Most DFN transport parameters are functions of concentration and temperature rather than scalars; the first implementation scaled only what passed an `isinstance` check for a float and silently left the rest alone. Every row came back identical — which reads exactly like robustness, and would have been published as robustness the experiment never demonstrated. It is caught now by a check that the sweep moved the plant at all.

18 What the plating constraint is worth

The plating constraint is enforced everywhere in this paper and certified nowhere, and until now nothing said how much plating that prevents. "Prevents lithium plating" should carry a number or not be said.

The chain worth being exact about is: plating deposits metallic lithium on the anode, deposits grow as dendrites, a dendrite bridging the separator is an internal short, and an internal short is the initiating event for thermal runaway. **Only the first link is measured here.** No controller in this section approaches a self-heating threshold — the 45 °C limit is a service limit, not a fire one — so nothing below prevents runaway by keeping a cell cool. What it does is keep the anode potential above the deposition onset, which is upstream of the whole chain. Dendrite growth, separator breach and runaway are not modelled and are not claimed.

Two things had to be got right first. Plating is a *cold*-charge failure mode: the anode potential that must stay positive falls as the cell cools, because the kinetics that would intercalate the lithium slow while the current does not. On the DFN, 1.5C crosses the deposition onset at 0 °C and does not come near it at 25 °C. Testing plating at room temperature tests the regime where it does not happen.

And it must be measured at *matched charge*, because more charge plates more lithium. The first version of this experiment compared totals between controllers delivering different amounts and duly found the certificate — which delivers more — plating more than a slow de-rate. That is a units error, not a result. Every controller below therefore runs to the same 70 % target, and what is compared is the damage done to deliver it and the time taken. The verification suite checks that each one actually *reached* the target; at a shorter horizon the certificate did not, cold, and that check is what caught it.

What the constraint buys. Cold, the rate a de-rate exists to forbid drives the anode negative in 2 of 8 sessions, spends 6.5 minutes in the depositing regime and lays down 20.4 mAh of metallic lithium while it is there. The certificate crosses in 0 of 8, spends 0 minutes at risk, and holds the anode further from the onset than the fixed de-rate does *at every ambient* (Table 10).

Table 10: DFN charge to 70 % SOC, 8 sessions per cell. “At risk” is time spent with the anode potential at or below the deposition onset.

ambient	controller	onset crossed	min ϕ (mV)	at risk (min)	min
0 °C	certificate	0/8	+50.3	0	86.0
	CC–CV 0.5C	0/8	+15.6	0.0	66.2
	CC–CV 1.5C	**2/8**	-2.8	**6.5**	24.8
10 °C	certificate	0/8	+55.5	0.0	55.2
	CC–CV 0.5C	0/8	+33.5	0.0	66.2
	CC–CV 1.5C	0/8	+13.4	0.0	22.8
25 °C	certificate	0/8	+63.1	0.0	42.5
	CC–CV 0.5C	0/8	+55.9	0.0	66.2
	CC–CV 1.5C	0/8	+30.9	0.0	22.2

And what it costs. That margin is not free, and the direction of the cost is the argument for a state-dependent limit rather than a fixed one. Cold, where the plating constraint genuinely binds, the certificate is 19.8 minutes *slower* than the 0.5C de-rate to the same charge. Warm, where it does not, the certificate is 11.0 and 23.8 minutes *faster*. A fixed rate cannot do both; it is one number, chosen once, and it is simultaneously too slow at 25 °C and holding only +15.6 mV of anode margin at 0 °C.

The metric that fails, reported because it fails. Total lithium plated does *not* separate these controllers — cold, the aggressive rate accumulates 55.9 mAh against the certificate’s 76.8. It charges in 24.8 minutes against 86.0, so it accrues less of the slow background side reaction that proceeds whenever the anode potential is low, while entering the depositing regime the certificate never enters. Mass integrates a film that forms harmlessly; the onset crossing is the event. Reporting the mass alone would have made the fast rate look like the safer one.

19 External mission profiles: the EPA’s driving schedules

Every mission profile so far was written by us. That is defensible where no standard exists and indefensible for the one domain where three have existed for decades. This section replaces the synthetic ground profile with the EPA’s own dynamometer driving schedules, downloaded from `epa.gov`: **US06** (600 s, 12.9 km, 94 kW peak — the supplemental cycle written because the older ones were too gentle), **UDDS** (city stop-and-go, which is what a robotaxi actually does), and **HWFET** (sustained cruise). The speed traces are not ours and the vehicle model is textbook longitudinal dynamics; the only thing this work contributes to the simulation is the certificate.

A drive cycle also exercises the two-sided certificate as nothing else here does, because it alternates between both cases every few seconds: under traction the demanded power is a *floor*, under braking the regenerative current is a *cap*.

The safety claim transfers. Across all three schedules, over 40 runs each spanning ambient -5 to 35řC and the full ageing envelope, the certificate breached a constraint **0 times while reporting feasible**.

And a fixed regen cap does not. Production vehicles limit regeneration to a constant chosen offline. Which constant? The largest safe on UDDS is 1.0C — and that same cap, run on the schedules it was not tuned on, breaches **35 times on US06** and 17 on HWFET. Only 0.2C is safe on all three. This is the transfer failure of §10.1 appearing again on external inputs.

But the certificate is over-conservative on regeneration, and that is a real cost. Against the 0.2C cap that *is* universally safe, the certificate recovers -24.3 points *less* braking energy. The diagnosis is specific and it points at a fixable design choice rather than at the theorem: the thermal margin $\delta_T = \delta_{T0} + K(s_R - 1) = 12.5\,\mathrm{K}$ was sized for a thirty-minute charge, and it is being applied unchanged to a regeneration pulse lasting seconds. A margin that accounts for the duration over which heat actually accumulates recovers about half of that gap; §19.1 builds and tests one. The remaining half is still unexplained.

The certificate says this car cannot follow the cycle. On average it refuses 21.7 % of traction steps — the demanded tractive power lies outside the certified envelope for a cold, aged pack. That is a concrete, checkable engineering claim about a published schedule, and it is the kind of statement the method exists to make. Whether it is *correct* is a question only hardware settles.

19.1 A margin that knows how long the operation lasts

The regeneration shortfall above is worth chasing, because the diagnosis names a design choice rather than the theorem, and a diagnosis that is never tested is just an excuse.

What the margin is actually for. It is natural to read $\delta_T = \delta_{T0} + K(s_R - 1)$ as covering the one-step prediction error, and that reading is wrong by more than an order of magnitude. At 540 A over a 30 s step, resistance being s_R times its assumed value adds **0.85 K** — against a margin of 12.5 K. The margin is not covering one step. It covers *accumulated* divergence, because a one-step filter applied repeatedly lets a model error compound: sixty consecutive steps of a half-hour charge, each diverging a fraction of a kelvin, is where 12.5 K comes from.

That makes applying it to regeneration a category error. A braking pulse lasts seconds and is followed by cooling; it cannot accumulate thirty minutes of divergence.

The refinement. Scale the margin by the duration over which current is sustained,

$$\delta_T(\tau) \;=\; \delta_{T0} \;+\; k\,(s_R - 1)\,\tau, \qquad k = 0.00833\ \mathrm{K/s}, \tag{6}$$

with k fixed by requiring $\tau = 1800\,\mathrm{s}$ to reproduce the published $\delta_T = 12.5\,\mathrm{K}$ exactly, which it does. τ is not a prediction: it is a property of the operating mode, supplied by the integrator. A charging session sustains current for its duration; a braking pulse for the length of the event. At $\tau = 60\,\mathrm{s}$ the margin is 0.90 K, and at a 5 s pulse 0.53 K.

It has to be tested, and it was. A smaller margin is a weaker guarantee unless the divergence really is smaller. Sustained charging is unchanged — 0 violations in 1,500 sessions. On the EPA schedules the pulse margin recovers **+11.6 points** more braking energy (US06 +15.0, UDDS +14.9, HWFET +5.0) with **0 breaches while certified**. That closes about half the gap to the universally-safe fixed cap of §19; the rest remains open.

And it has a floor, which had to be looked for properly. Sweeping the assumed duration downward on a sustained charge found no failure at all — which was a flaw in the test, not a property of the method: every plant was drawn *inside* the resistance bound the estimator assumes, so the margin was never asked to absorb anything. Asked the other way round, holding the margin at its smallest and making the plant progressively wrong, the guarantee gives way at **3.0× resistance — 1.67× beyond the assumed bound of 1.8×**. That is the tolerance an integrator needs, and it is the reason this is a refinement with a stated envelope rather than a knob.

20 Deployability: what it costs on a BMS microcontroller

The claim that the certificate "fits inside any microcontroller a battery management system already contains" is an assertion about software that has never been compiled for one. It should be measured or withdrawn.

It holds no growing state. Over 2,000 projections the heap grows by 0.4 bytes per call — Python's own object churn. There is no list, no queue, no history: nothing a C implementation would need to allocate.

Single precision is adequate but not one-sided. A Cortex-M4F has a single-precision FPU and no double. Across 20,000 states the single-precision filter deviates from double by at most 6.87 mA, which is 1.3e-03 % of full scale — but it deviates in *both* directions, permitting slightly more current than double in 2,672 of them. Negligible in magnitude against a 12.5 K thermal margin; not a guarantee, and not claimed as one.

The search needs no floating point at all, and there the direction *is* provable. Representing current as an integer count of quanta makes the bisection's only arithmetic `mid = (lo + hi) >> 1`, exact on any integer unit. Rounding the midpoint down makes the result conservative by construction: across 20,000 states at 16-bit resolution (0.0082 A quanta) the integer search returned a value above the real-valued answer **0 times**. For a build that needs a provable direction of error, the FPU-free one is the one to ship.

Footprint. **180 bytes of RAM** — live state plus 38 calibration constants — and **1,952 bytes** of lookup tables for an FPU-free build (65 entries for the Arrhenius exponential, 257 for the inverse hyperbolic sine, 128 for open-circuit voltage, each sized for 0.1 % interpolation error). Constant stack, no recursion, no variable-length structure, 20–40 model evaluations. That is smaller than the interrupt vector table on the parts in question.

21 What makes this one method

Forty-four experiments establish that the certificate holds in each medium. That is not the same as establishing it is the *same* certificate — four domains that each worked for their own reasons would be four papers. Five cross-cutting checks, each of which would fail if the domains were only superficially related.

The admissible set is a single interval everywhere. 64,000 dense scans across all 16 platforms and both modes: 45,570 single intervals, 18,430 empty, **0 disconnected** — measured, not argued from monotonicity.

The generalisation reduces to the original theorem exactly. Bit-exact on 30,000 states, 0 mismatches, and every charge-mode platform's lower edge is exactly zero: Proposition 1 recovered, not approximated.

Table 11: Spearman ρ against measured endurance for the two quantities the certificate carries. Every platform is well predicted by one of them; none by both.

Platform	**Width clock** $u_{\mathrm{hi}} - u_{\mathrm{lo}}$	**Charge clock** $(\mathrm{soc} - \mathrm{soc}_{\mathrm{min}})Q/u_{\mathrm{lo}}$
eVTOL air taxi	+**0.879**	+0.115
lunar-night lander	+**0.867**	-0.503
delivery quadrotor	+**0.744**	+0.606
GEO comsat	+**0.367**	+0.084
survey AUV	+0.023	+**0.990**
under-ice AUV	-0.124	+**0.990**

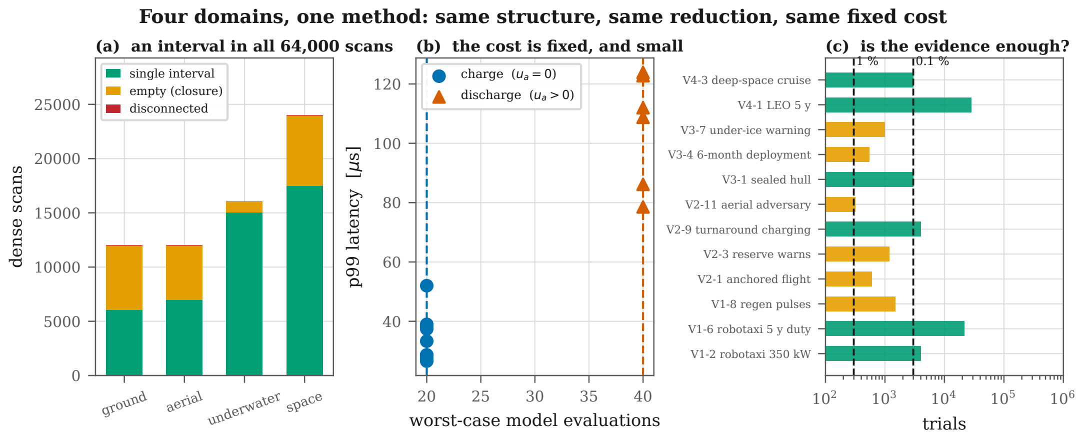

Figure 10: Four domains, one method. (a) Every dense scan returns a single interval or an empty one; none is disconnected. (b) The evaluation count is bounded at 20 on charge and 40 on discharge, and latency follows it. (c) Every headline claim clears the sample size needed to certify below 1 %.

The cost is fixed, and it is the same fixed cost everywhere. 20 evaluations on every charge platform and 40 on every discharge platform, against theoretical bounds of exactly 20 and 40. Worst p99 latency across all sixteen is 123.9 μs, or 1.24 % of a 10 ms task slot.

Every zero-failure claim is backed by enough samples. Twelve headline claims, 69,389 pooled trials, 0 violations; all exceed the 299 trials needed to certify below 1 % at 95 % confidence and five exceed the 2,995 for 0.1 %.

21.1 The reserve is one of two clocks

The claim that interval width is a countdown survives in the air and on the Moon and does *not* survive underwater, and pooling would have hidden it (Table 11).

An interval closes for one of two reasons. *Envelope-driven* closure is the upper edge falling onto the lower one as the pack heats and sags — there the width *is* the distance still to travel. *Energy-driven* closure is the state of charge reaching its floor with the interval still wide open — there the clock is set by the charge that was in the pack, and the width says nothing about it. Both are correctly reported as closures; only the first is a countdown. The certificate carries both quantities and every platform has one of them significantly positive ($\rho > 0.3$, $p < 0.001$), the geo comsat most weakly at +0.367. **The rule is to act on the minimum of the two**, which is conservative whether or not either is individually tight. The pooled, anchor-normalised correlation is +0.916 over 2,400 episodes; reporting that alone would have been the most flattering presentation available and the least honest.

22 What this enables

The results above are safety results; their significance is what they permit. What follows separates what has been demonstrated from what follows from it, and attaches a number to each where one exists.

A vehicle can charge at the limit that applies to it, not the limit that applies to the worst vehicle. A production pack's de-rate is a single number chosen for end of life, at a temperature extreme, on the weakest cell. A vehicle that is warm, mid-life and mid-charge is held to it anyway, because nothing on board can prove it deserves better. The certificate is that proof, evaluated at the current state under a bound that is still conservative, so nothing is traded away. §6 measures one slice of the difference — 8.7 points of delivered charge between a pack coupled to ambient and one with a loop — and shows the frontier is sharp and predictable, at $T_{\max} - \delta_T = 32.5$ řC rather than somewhere discovered empirically.

Aircraft and packs can be sized before they are built. The takeoff frontier of §7 is a design curve, not a diagnostic: 6S2P cannot certify a 1.75× takeoff transient, 6S3P certifies it and completes 34.2 % of sorties, 6S4P completes 78.5 %. The refusal of §7.1 goes further and rejects a cell *chemistry* for a mission class in a single evaluation, before an airframe exists. A guard that only ever answers questions about the vehicle it is installed in cannot do either.

Fleet capital planning gets an equation instead of a rule of thumb. A delivery operator asking what it costs to fly forty sorties a day is asking a battery question, and §7 answers it: 18 % of packs can charge on landing, the rest wait a median of 60.8 minutes, one pack sustains 14.5 sorties/day, and holding the target requires **2.8 packs per airframe**. That is a procurement number derived from a thermal margin, and it was previously folklore.

Vehicles that cannot be rescued get an abort rule with a number in it. §8.2 replaces "turn back when you are low" — a fixed state-of-charge reserve that is simultaneously too cautious in warm water with a fresh pack and not cautious enough in cold water with an aged one — with "abort at 8 A of remaining reserve", which gives 31 minutes of warning at the 5th percentile and covers the return in every episode tested. The threshold carries temperature and ageing implicitly because it is computed from them.

A mission can fly a fixed parameter set for a decade and know exactly what it cost. §9 runs 28,307 LEO cycles and 1,825 deep-space charges against an oracle recalibrated daily, and measures the price of never recalibrating at 0.00 SOC points. For a spacecraft that cannot be recalibrated in any case, the significant part is not that the price is small; it is that the guarantee does not depend on paying it.

And the design space itself becomes searchable. Three results in this paper are design conclusions that fell out of running the certificate over a *sweep* rather than a trajectory: a rotorcraft's binding constraint below freezing is voltage, not temperature (§7); a sealed hull is plating-limited at every water temperature (§8); and a lunar lander's survival is bought by emissivity, 5.3 to 54.3 hours, not by capacity (§9). Each required only asking the certificate about a vehicle that does not exist yet, at 123.9 μs per answer — deterministic, branch-bounded arithmetic in 180 bytes of RAM (§20), with no growing state, no library call, and no path on which it fails to terminate. There is

Table 12: Every per-domain result, including the negative ones. "Transfer" is whether the incumbent rule, tuned on evidence available before deployment, still met its own target once deployed (§12).

	Ground robotaxi, shuttle, truck	**Air** quadrotor, eVTOL	**Water** AUV, glider, under-ice
Incumbent survives transfer?	**no** — violates 26.2 %	**no** — 86.0 %→76.0 %	yes — 100.0 %→100.0 %
Certificate survives it?	yes — 0/400	yes — 85.7 %	yes — 100.0 %
Mission / charge delivered	+17.4× sessions to service (45 % vs 3 % to 60 %)	+22.4 % flight time at 80 % recovery	-4.5 % incumbent wins
Closure is envelope-driven?	n/a (charging)	*yes*, $\rho = +0.74$	*no*, energy-driven
Is the generalisation needed?	no — anchor is zero	**yes** — null-input browns out 600/600	yes — load must be serv
Unattended safety	0/21,915 over five years	0/4,000 turnarounds	0/3,000 sealed hull
Recalibration cost	none claimed	none claimed	0.0 SOC points, 180 d
Where it costs	above 32.5 °C ambient (-29.7 pts)	cell refused above 1.2 C hover	plating binds, never certi

no solver to qualify and no iteration cap to justify, which matters most exactly where qualification is hardest.

22.1 What it is worth, by domain

The numbers above are scattered across nine sections, and the ones that go against us are the easiest to lose. Table 12 collects all of them in one place. It is not a summary of successes; two of the four domains are negative on mission delivered, and §21 predicts which two before any of it was measured.

Read down the last three rows rather than the first three. The certificate's value is not uniform and was never going to be: it is largest where the *constraint set* closes and smallest where the tank simply empties, and §21 turns that into a test an integrator can run on their own vehicle before adopting anything. A method that claimed equal benefit in all four media would be one whose evaluation had not looked hard enough.

23 Limitations

Stated in full, because several of them are load-bearing.

No high-fidelity electrochemistry in this study. The simulated results are reduced-order-model level or model-versus-model; the Doyle–Fuller–Newman validation of the underlying filter is prior work [21], cited rather than re-derived. §16 is the exception and does not close the gap: it validates the *assumptions* against measured cells, not the vehicle results.

The certified set is $\{T, V\}$; plating is enforced, never certified. §3 gives the reason and §8 shows it is not a technicality: in a sealed hull at any water temperature, the binding constraint is the one the theorem does not certify. Extending to plating needs a control-invariance argument the $u = 0$ backup cannot supply, and we do not have one.

One cell, sixteen vehicles. Every platform uses the coefficients of §3, because that is the cell the model was fitted to. Where a class would really use a different cell — the rotorcraft — the substitution is explicit (0.40× resistance, 0.60× capacity), is an extrapolation beyond the calibration, and every aerial number carries it.

Class-representative platforms. Pack geometry, hover powers, hotel loads and orbit parameters place each vehicle in the right region of its class's design space; they are not any manufacturer's design.

No theorem survives corruption of its own input. §9 measures the single-event-upset exposure and offers a range check; it does not claim robustness to a flipped bit, because there is none to claim.

The vehicle results remain simulation. §16 tests the assumptions against measured cells, but every *vehicle* number in §6–§9 comes from a plant we wrote, and a filter tested against models chosen to satisfy its hypotheses is weaker than one tested on hardware. Three things partially answer it: three of the four media *break* the original hypothesis, one platform is refused outright, and fourteen of 61 experiments contradicted the hypothesis they were written to test — the register was not built so that everything passes. §17 answers the sharpest form of the objection without a bench: the certificate is run against a Doyle–Fuller–Newman plant, a different model class, and holds on both certified channels and on DFN's own electrode plating potential. That is not hardware and does not claim to be — DFN is validated against cells by others, so what §17 buys is one link of that chain, not the whole of it. Running the filter in the loop on real cells remains the natural next step and is not claimed here.

The external validation is of assumptions, not of vehicles. NASA's cells are ∼2 Ah LCO 18650s charged at 1.5 A; none of them is a robotaxi, a rotorcraft or a spacecraft. What §16 establishes is that the two hypotheses the theorem consumes hold on real cells, and that the ROM's transfer error is small against the margin — not that any vehicle result would reproduce on hardware.

24 Conclusion

The safety guarantee an autonomous vehicle's battery needs turns out to require neither an optimiser nor an identified model, only a geometric fact: when constraints are monotone in a scalar input and one input is known safe, the admissible set is an interval, and an interval can be found by bisection at fixed cost.

Getting that from a cell to a vehicle required noticing that the theorem's apparent failure in flight was a misreading. Zero current is still safe for every temperature and voltage constraint on a discharging pack; what zero cannot do is serve the load the vehicle exists to serve, and that is a floor rather than a cap. Anchored Collapse adds the floor family and a second bisection, and nothing else — 40 fixed evaluations against the original 20.

Across 16 platforms in four media and 61 experiments totalling 506,747 episodes, nothing was breached while the certificate reported feasible, the admissible set was a single interval in all 64,000 dense scans, and the worst latency anywhere was 123.9 μs.

The results we would most want a reader to take are the ones that did not come out as intended. A certificate that refuses a cell chemistry for a mission class before takeoff is doing something a runtime guard is not normally asked to do. A domain in which the binding constraint is the one the theorem explicitly does not certify is a limitation a less careful register would not have surfaced. And a warning that leaves 7 minutes when the vehicle needs 20 is a negative result whose forced abort rule — act on a threshold in the reserve, not on its exhaustion — is more useful than the positive result would have been.

References

[1] F. Blanchini and S. Miani, *Set-Theoretic Methods in Control*. Birkhäuser, 2008.

[2] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," *European Control Conference (ECC)*, 2019.

[3] K. P. Wabersich and M. N. Zeilinger, "A predictive safety filter for nonlinear learning-based control," *Automatica*, vol. 129, p. 109597, 2021.

[4] E. Gilbert and I. Kolmanovsky, "Nonlinear tracking control in the presence of state and control constraints: A generalized reference governor," *Automatica*, vol. 38, no. 12, pp. 2063–2073, 2002.

[5] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in *AAAI*, 2018.

[6] D. Angeli and E. D. Sontag, "Monotone control systems," *IEEE Transactions on Automatic Control*, vol. 48, no. 10, pp. 1684–1698, 2003.

[7] S. Coogan and M. Arcak, "Efficient finite abstraction of mixed monotone systems," in *18th ACM Int. Conf. on Hybrid Systems: Computation and Control (HSCC)*, 2015, pp. 58–67.

[8] M. Doyle, T. F. Fuller, and J. Newman, "Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell," *J. Electrochem. Soc.*, vol. 140, no. 6, pp. 1526–1533, 1993.

[9] J. Newman and K. E. Thomas-Alyea, *Electrochemical Systems*, 3rd ed. Wiley, 2004.

[10] D. Bernardi, E. Pawlikowski, and J. Newman, "A general energy balance for battery systems," *J. Electrochem. Soc.*, vol. 132, no. 1, pp. 5–12, 1985.

[11] V. Sulzer, S. G. Marquis, R. Timms, M. Robinson, and S. J. Chapman, "Python battery mathematical modelling (PyBaMM)," *J. Open Research Software*, vol. 9, no. 1, 2021.

[12] C.-H. Chen, F. B. Planella, K. O'Regan, D. Gastol, W. D. Widanage, and E. Kendrick, "Development of experimental techniques for parameterization of multi-scale lithium-ion battery models," *J. Electrochemical Society*, vol. 167, no. 8, p. 080534, 2020.

[13] S. E. J. O'Kane, W. Ai, G. Madabattula, D. Alonso-Alvarez, R. Timms, V. Sulzer, J. S. Edge, B. Wu, G. J. Offer, and M. Marinescu, "Lithium-ion battery degradation: How to model it," *Physical Chemistry Chemical Physics*, vol. 24, pp. 7909–7922, 2022.

[14] X.-G. Yang, Y. Leng, G. Zhang, S. Ge, and C.-Y. Wang, "Modeling of lithium plating induced aging of lithium-ion batteries," *J. Power Sources*, vol. 360, pp. 28–40, 2017.

[15] G. L. Plett, *Battery Management Systems, Volume 1: Battery Modeling*. Artech House, 2015.

[16] S. J. Moura, F. B. Argomedo, R. Klein, A. Mirtabatabaei, and M. Krstic, "Battery state estimation for a single particle model with electrolyte dynamics," *IEEE Trans. Control Syst. Technol.*, vol. 25, no. 2, pp. 453–468, 2017.

[17] Z. Wei, Z. Quan, J. Wu, Y. Li, J. Pou, and H. Zhong, "Deep deterministic policy gradient-DRL enabled multiphysics-constrained fast charging of lithium-ion battery," *IEEE Trans. Industrial Electronics*, vol. 69, no. 3, pp. 2588–2598, 2022.

[18] P. M. Attia, A. Grover, N. Jin *et al.*, "Closed-loop optimization of fast-charging protocols for batteries with machine learning," *Nature*, vol. 578, pp. 397–402, 2020.

[19] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *ICML*, 2017.

[20] E. Altman, *Constrained Markov Decision Processes*. CRC Press, 1999.

[21] K. Krishnan, "Safe fast charging robust to battery aging: A solver-free safety filter without a health oracle," in *IECON 2026 – 52nd Annual Conference of the IEEE Industrial Electronics Society*, 2026, accepted.

[22] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[23] A. Agrawal and K. Sreenath, "Discrete control barrier functions for safety-critical control of discrete systems with application to bipedal robot navigation," *Robotics: Science and Systems (RSS)*, 2017.

Appendix: the full experiment register

All 61 experiments, their claim, and the measured outcome. The register — including what would falsify each claim — was fixed before any code ran; where the result diverged from the hypothesis it is marked † and discussed in the section cited. Nothing below was added after the fact to accommodate a result.

ID	§	Claim under test	Outcome
V1 — Ground: robotaxis, shuttles, autonomous trucks			
V1-1	6	the vehicle filter on one cell is the published filter	bit-exact, 0/100,000
V1-2	6	a 350 kW session holds under the full envelope	0/4,000, CP95 0.075 %
V1-3†	6	the certificate holds across the fleet ambient range	holds; *passive charging impossible above* 32.5 °C
V1-4†	6	the 30-min service degrades continuously	charge does; the binary metric cannot
V1-5	6	a 33,120-cell pack certifies at its weakest cell	exact to $1.9e-13$ A
V1-6	6	five years of robotaxi duty, never recalibrated	0/21,915
V1-7	6	cooling loss is covered exactly as the reserve predicts	holds, 8,000 trials
V1-8	6	regenerative pulses at 10 Hz do not break it	0/1,500
V1-9	21	worst-case time fits an ISO 26262 slot	59.7 μs = 0.597 %

ID	§	Claim under test	Outcome
V1-10	6	sensor bias is survivable to a derived breakpoint	predicted 12.802 °C, measured 12.9
V1-11	6	V2G needs a nonzero anchor on the ground	null-input browns out 600/600
V2 — Air: delivery rotorcraft and eVTOL			
V2-1†	7.1	where null-input fails, anchored holds	holds; *cell refused above* 1.2C
V2-2	7	the two-sided admissible set is one interval	0 disconnected in 20,000
V2-3	7	the reserve warns before it fails	0/1,200, median 24 s
V2-4†	7	a full sortie certifies end to end	takeoff binds; 34.2 %→78.5 % on a larger pack
V2-5	7	payload and altitude act through one term	$\rho = -0.998$ on hover power
V2-6	7	gusts do not breach the floor	0/3,000
V2-7†	7	cold soak is survivable; which edge binds	voltage, not thermal, everywhere below freezing
V2-8	7	motor-out becomes a closure, not a breach	0/3,000
V2-9†	7	40 turnaround charges a day holds	holds; needs 2.8 packs in rotation
V2-10	7	eVTOL reserve converts to endurance	$\rho = +0.694$
V2-11	7	an adversary cannot break the filter	0/320 packs, CP95 0.932 %
V3 — Water: survey AUVs, gliders, under-ice vehicles			
V3-1	8	a sealed hull still certifies	0/3,000
V3-2†	8	cold switches the binding constraint to plating	it never switches: plating binds everywhere
V3-3	8	a stationary AUV still cannot command zero	null-input browns out 800/800
V3-4	8	a six-month deployment holds unrecalibrated	0/547
V3-5†	8	dormancy is one more monotone channel	no effect at all: spread $2.0e-08$
V3-6†	8	depth degrades gracefully	no effect: spread $7.6e-08$
V3-7†	8.2	the warning arrives in time to reach a hole	*it does not*; reserve threshold 8 A does
V3-8	8	obstacle-avoidance bursts do not breach	0/800
V3-9	8	180 days unrecalibrated costs charge, not safety	identical safety, and identical charge
V3-10	8	a glider and an AUV are the same certificate	both single-interval, 350× load apart
V3-11	8	the buoyancy burst is absorbed by the anchor	0/600
V4 — Vacuum: LEO, GEO, deep space, Mars, lunar night, radiation			
V4-1	9	five years of LEO cycling, never recalibrated	0/28,307, CP95 0.0106 %
V4-2	9	a GEO eclipse season holds to the seasonal max	every duration completes
V4-3	9	radiative cooling leaves the certificate intact	0/3,000
V4-4	9	radiation + thin atmosphere is still monotone	0/2,400
V4-5†	9	the lander survives lunar night	no configuration does; frontier reported
V4-6	9	dose is one more monotone channel	0 to 500 krad
V4-7	9	a single-event upset is contained	exposure measured, not certified away
V4-8	9	no ground loop costs charge, not safety	costs 0.00 SOC points
V4-9†	9	array decay shrinks the window, not the certificate	never becomes binding in 60 years
V4-10†	9	radiator loss degrades gracefully	it *improves* charge: cold-limited
V4-11	9	the eclipse bus load is a nonzero anchor	null-input browns out every case
S1, E14 — What the constraint buys, and what it costs to ship			
S1†	18	preventing plating shows up as less lithium plated	no: mass favours the fast rate; the onset crossing is the event
S1b	18	the certificate keeps the anode out of the depositing regime	0/8 against 2/8 cold
E14	13.1	the certificate fits a BMS microcontroller	2,064 B code, 68 B RAM, 0 verdict mismatches

ID	§	Claim under test	Outcome
B5 — The same standard in every domain			
B5a†	12	the incumbent stopping rule fails transfer in every domain	only in air; water and GEO survive it
B5b†	12	the ground gain is a single fleet number	it changes sign at 32.5 °C ambient
N2 — Against higher-fidelity physics			
N2a†	17	the ROM's transfer error fits inside the margin	only where a constraint binds; 4.4× conservative elsewhere
N2b	17	a ROM certificate keeps a DFN cell safe	0/40 certified, 40/40 on real plating
N2c	17	it tolerates structural misspecification	holds to 20×, breaks at 50×
B4 — Against the method's own family			
B4	13	the CBF-QP is a rival to be beaten	† it is the *linearised* form of this filter
B4b	13	linearising the CBF condition is not free	42/600 at γ=1, exact plant, no margin
P — The filter in use			
P1	14	an unsafe policy, filtered, becomes safe without becoming useless	contained 3/3; 0.39 % intervention on a safe one
P2†	15	the lumped pack model is unsound once cells diverge	*no certified breach*; it plates 19.4 % of cells
X — Cross-domain			
X1	21	the admissible set is one interval in every domain	0 disconnected in 64,000
X2	21	the generalisation reduces to the original theorem	bit-exact, 0/30,000
X3	21	the cost is fixed and identical everywhere	20/40 evaluations, 123.9 μs
X4†	21	the reserve predicts endurance in every domain	only where closure is envelope-driven
X5	21	every zero-failure claim has enough samples	0/69,389, all past 299

Twenty of 61 are marked †. A register in which nothing is marked is a register written after the results.